Looking again at figure 12.8, the specificities of the T-cells entering the periphery from the thymus must be moulded by the self-peptides, which drive positive selection, since normally the only peptides around must be derived from self. It is satisfying to note therefore that the T-cell repertoire tends to be biased towards peptides from extrinsic antigens which resemble self; thus, T-cell epitopes recognized on immunization with xenogeneic lysozyme corresponded with sequences having the highest homology to the syngeneic protein.

T-cell tolerance can also be due to clonal anergy

We have already entertained the idea that engagement of the TCR plus a costimulatory signal from an antigen-presenting cell are both required for T-cell stimulation, but, when the costimulatory signal is lacking, the T-cell becomes tolerized or anergic, or, if you prefer, paralysed.

Thus, anergy can be induced in **extrathymic T-cells** by peripheral antigens *in vivo* when presented by cells lacking costimulatory molecules. If a transgene construct of H-2E^b attached to an insulin promoter is introduced into a mouse which normally fails to express H-2E, the H-2E^b transgene product appears on the β-cells of the pancreas and induces tolerance to itself. Whereas the expression of H-2E on bone marrow-derived cells in the thymic medulla deletes T-cells bearing Vβ17a receptors, these cells are not lost in the tolerant transgenic mouse expressing pancreatic H-2E, i.e. there is a state of clonal anergy, not deletion. The altered immunological status of these cells is revealed by their inability to proliferate when their receptors are cross-linked by an antibody to Vβ17a.

It is unlikely that these results are due to low level expression of antigen in the thymus. Similar experiments showed that mice expressing influenza hemagglutinin on the pancreatic β-islets also became tolerant irrespective of whether the transgenic thymus was replaced by a normal gland or not. Nonetheless, anergic cells can also be generated within the thymic population as seen in mice transgenic for both an anti-K^b TCR and a K^b gene controlled by a truncated fragment of a keratin IV promoter which allowed expression on thymic medullary cells.

Peripheral T-cell anergy can occur at different levels depending upon the circumstances of antigen exposure. If the above double transgenic experiment is repeated with a full keratin IV promoter, the K^b antigen is expressed on keratinocytes and induces full tolerance, even though the same high frequency of cytotoxic T-cell precursors with the transgene TCR is seen as in single transgenic animals lacking K^b. If K^b is expressed on cells of neuroectodermal origin or hepatocytes, again the double transgenic mice are tolerant but there is dramatic downregulation of TCR and CD8 molecules; in the former but not the latter case, downregulation of TCR could be reversed by exposure to antigen *in vitro*. In some experimental models, tolerance can be abrogated by IL-2. To recapitulate, autoreactive T-cells leaving the thymus can be rendered anergic in the periphery and can display different degrees of potentially reversible unresponsiveness.

Infectious anergy

If a clone of T-helpers is subject to a limiting dilution experiment (p. 137), the minimal unit of proliferation in response to peptide on an APC is usually several cells not just one. This implies that triggering only occurs in small groups or clusters of cells and suggests that paracrine or multicellular interactions between potential responders bound to a single APC are needed to drive the cells into division (figure 12.9a). It will be appreciated that, if a newly arising extrathymic naive T-cell binds to its antigen, even on a professional APC, it will not be stimulated if its neighbors in the cluster have already been made anergic. Indeed, instead of being triggered, it will itself become anergic, so perpetuating the infectious anergic process (figure 12.9b). Recent evidence suggests that anergic T-cells can act as immunoregulatory cells by causing the downregulation of MHC class II and the costimulatory CD80 (B7.1) and CD86 (B7.2) molecules on the APC, generating an infectious anergy which does not require the simultaneous presence of the regulatory anergic cells and the cells which will themselves be made anergic (figure 12.9c). We shall see later in Chapter 17 that the induction of transplantation immunosuppression with a nondepleting anti-CD4 can be long-lasting because the production of anergic cells prevents the priming of newly immunocompetent T-lymphocytes by the transplantation antigen(s).

These anergic cells are really acting as suppressors. So far in our discussions, we have not asked the question: do dedicated T-suppressors contribute to self-tolerance? Frankly, another gray area, but experimentally we can demonstrate that, if autoimmunity is induced in a normal animal, either actively, by injection of an antigen cross-reacting with self, or passively, by injection of autoreactive T-cells (cf. p. 416 and p. 433), the self-reacting clones are usually quashed by idiotype- or antigen-specific T-suppression. In a nutshell, suppressors probably do not prevent autoimmunity but they may reverse it.

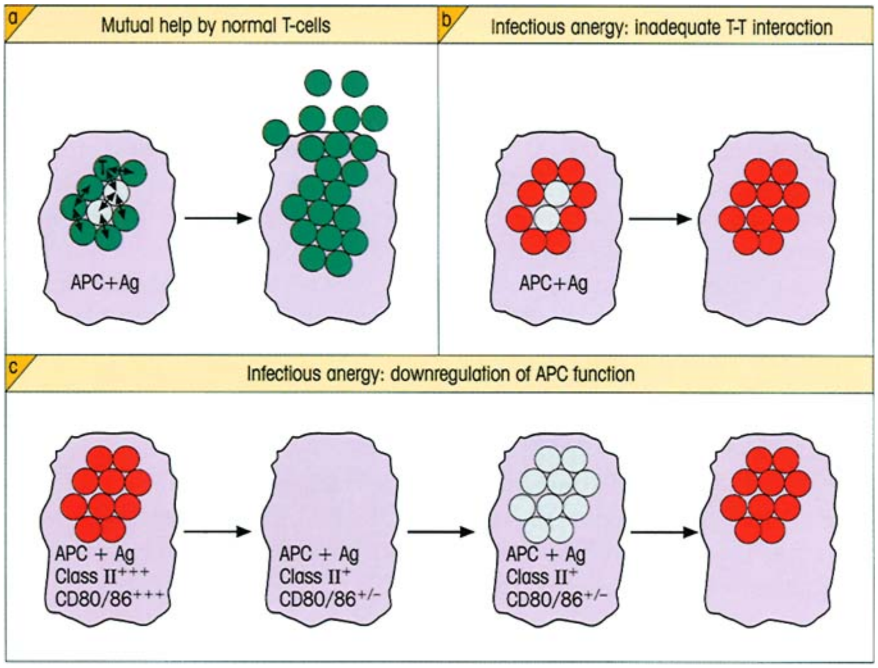

Figure 12.9. Infectious anergy. (a) Clusters of normal T-cells (green) around newly immunocompetent cells (gray) reacting with the same APC mutually support activation and proliferation. (b) Newly immunocompetent cells surrounded by anergic T-cells (red) receive no stimulatory signals from their neighbors and are themselves rendered anergic. (c) Upon being made anergic, T-cells can assume an immunoregulatory function whereby they downregulate expression of MHC class II, CD80 (B7.1) and CD86 (B7.2) molecules on the APC. This effect requires cell–cell contact between the anergic T-cells and the APC, is not blocked by neutralizing antibodies to the cytokines IL-4, IL-10 or TGFβ, and would exhibit linked suppression to other epitopes on the antigen presented by the APC. Subsequent encounter of newly immunocompetent cells with this APC will lead to anergy.

Lack of communication can cause unresponsiveness

It takes two to tango: if the self-molecule cannot engage the TCR, there can be no response. The anatomical isolation of molecules, like the lens protein of the eye and myelin basic protein in the brain, virtually precludes them from contact with lymphocytes, except perhaps for minute amounts of breakdown metabolic products which leak out and may be taken up by antigen-presenting cells, but at concentrations way below that required to trigger the corresponding naive T-cell.

Even when a tissue is exposed to circulating lymphocytes, the concentration of processed peptide on the cell surface may be insufficient to attract attention from a potentially autoreactive cell in the absence of costimulatory B7. This was demonstrated rather elegantly in animals bearing two transgenes: one for the TCR of a CD8 cytotoxic T-cell specific for LCM virus glycoprotein, and the other for the glycoprotein itself expressed on pancreatic β-cells through the insulin promoter. The result? A deafening silence: the T-cells were not deleted or tolerized, nor were the β-cells attacked. If these mice were then infected with LCM virus, the naive transgenic T-cells were presented with adequate concentrations of the processed glycoprotein within the adjuvant context of a true infection and were now stimulated. Their *primed* progeny, having an increased avidity (cf. p. 388) and thereby being able to recognize the low concentrations of processed glycoprotein on the β-cells, attacked their targets even in the absence of B7 and caused diabetes (figure 12.10). This may sound a trifle tortuous, but the principle could have important implications for the induction of autoimmunity by cross-reacting T-cell epitopes (cf. p. 412).

Molecules that are specifically restricted to particular organs which do not normally express MHC class II represent another special case, since they would not have the opportunity clonally to delete or paralyse organ-specific CD4 T-helper cells.

Immunological silence would also result if an individual has no genes coding for lymphocyte receptors directed against particular self-determinants; analysis of the experimentally induced autoantibody response to cytochrome suggests that only those parts of the molecule which show species variation are autoantigenic, whereas the highly conserved regions where the genes have not altered for a much longer time appear to be silent, supposedly because the autoreactive specificities have had time to disappear.

B-CELLS DIFFERENTIATE IN THE FETAL LIVER AND THEN IN BONE MARROW

The B-lymphocyte precursors, pro-B-cells, are present among the islands of hematopoietic cells in fetal liver by 8–9 weeks of gestation in humans and 14 days in the mouse. Production of B-cells by the liver wanes and is mostly taken over by the bone marrow for the remain-

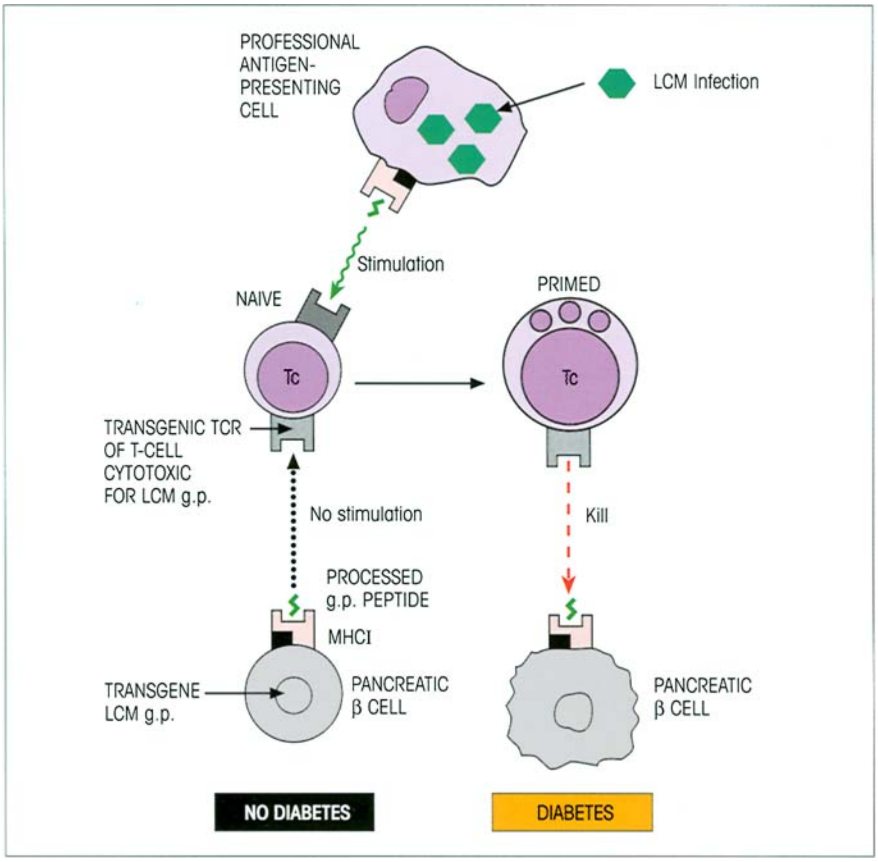

Figure 12.10. Mutual unawareness of a naive cytotoxic precursor T-cell and its B7-negative cellular target bearing epitopes present at low concentrations. Priming of the naive cell by a natural infection and subsequent attack by the higher avidity primed cells on the target tissue. LCM, lymphocytic choriomeningitis virus. (From Ohashi *et al.* (1991) *Cell* **65**, 305.)

der of life. Using the modified culture conditions introduced by Whitlock and Witte, it is now possible to grow bone marrow cells *in vitro* and achieve the differentiation of B-cells and their precursors. Stromal reticular cells, which express adhesion molecules and secrete IL-7, extend long dendritic processes making intimate contact with IL-7 receptor-positive B-cell progenitors. Although early B-cells comprise only a minor subpopulation of the cells in those cultures, it is possible to analyse the different stages in their development by rescue with the Abelson murine leukemia virus (A-MuLV), a replication-defective retrovirus capable of transforming pre-B-cells at various points in their development into clones. A series of differentiation markers associated with B-cell maturation have now been established (figure 12.11).

Pax5 is a major determining factor in B-cell differentiation

Development of hematopoietic cells along the B-cell lineage requires expression of E2A and of early B-cell factor (EBF); the absence of either of these prevents pro-B-cells progressing to the pre-B-cell stage (figure 12.12). Also required is expression of the *Pax5* gene which encodes the BSAP (*B*-cell-*s*pecific *a*ctivator *p*rotein) transcription factor. Thus, in *Pax5*$^{-/-}$ knockout mice, early pre-B-cells (containing partially rearranged immunoglobulin heavy chain genes) fail to differentiate into mature, surface Ig^{+}, B-cells (figure 12.12). However, if the pre-B-cells from *Pax5*$^{-/-}$ knockout mice are provided with the appropriate cytokines *in vitro*, they can be driven to produce T-cells, NK cells, macrophages, dendritic cells, granulocytes and even osteoclasts! These unexpected findings clearly show that the early pre-B-cell has the potential to be diverted from its chosen path and instead provide a source of cells for many other hematopoietic lineages. However, these pre-B-cells are not pluripotent as, unlike bone marrow stem cells, they are unable to rescue lethally irradiated mice. In the light of these findings, it has been proposed that *Pax5* acts as a master gene critical for B-cell development and functions by inhibiting, rather than activating, the expression of a set of genes. *Pax5* expression thereby suppresses alternative lineage choice in early B-cells.

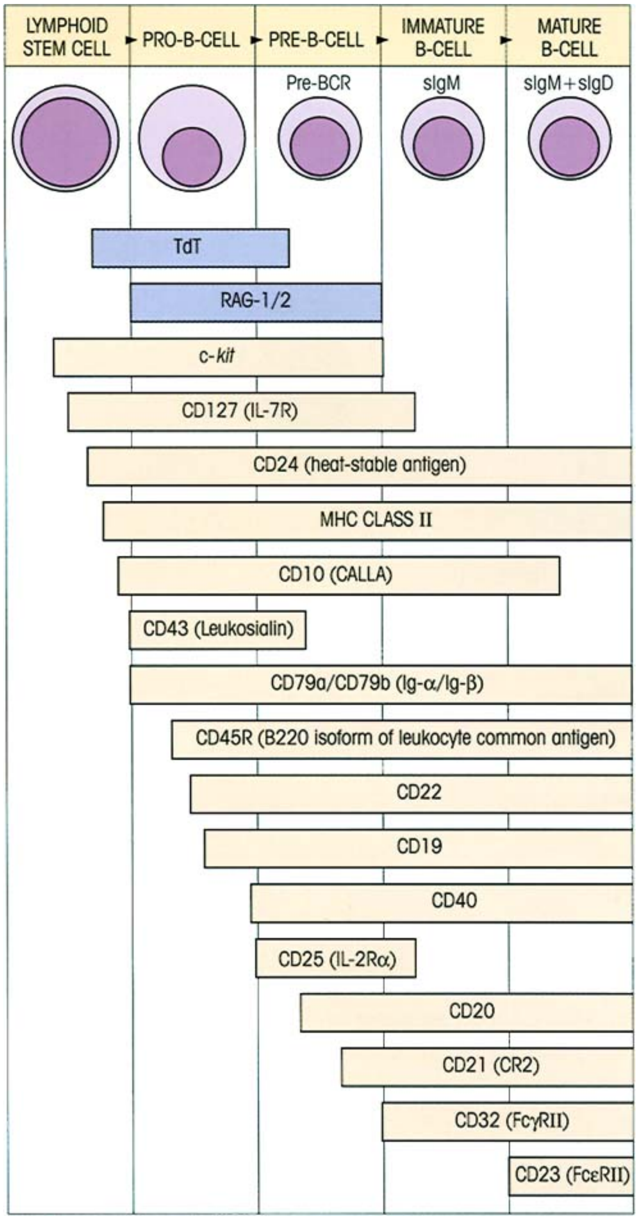

Figure 12.11. Some of the differentiation markers of developing B-cells. The time of appearance of enzymes involved in Ig gene rearrangement and diversification (blue boxes) and of surface markers defined by monoclonal antibodies (orange boxes, see table 8.1 and Appendix 1 for list of CD members) is shown.

Table 12.3. Comparison of two mouse B-cell subsets. (Developed from Herzenberg L.A., Stall A., Melchers F. *et al.* (eds) (1989) *Progress in Immunology* 7, p. 409. Springer-Verlag, Berlin.)

	B-1	B-2
PHENOTYPE		
IgM	+++	+
IgD	+	+++
CD5	+ or –	–
CD23	–	+
CD43	+	–
MAIN LOCATION	Peritoneal cavity	Lymphoid organs
ONTOGENY	Arise first in fetal liver	Arise later in adult bone marrow
LIFESPAN	Self-renewing Constitutive production IL-10	Replaced by IgM^- precursors in bone marrow
GROWTH	Propensity to expansion	Die easily
IMPAIRED DEVELOPMENT	Xid (CBA/N)[1]	me^v (motheaten)[2]
Ig GENES	Unmutated, little N-nucleotide insertion	Mutated, common N-nucleotide insertion
ANTIBODY PRODUCTION		
Serum IgM, IgG3	+++	+
IgG1	+	+++
IgG2a, IgG2b	+ to ++	++ to +++
IgM autoantibody	+++	?
IgM anti-Id	+++	?
IgM anti-bacterial Ab	+++	+ to +++
Anti-hapten/protein	?	+++
T-dependence	–	++
Affinity maturation	–	++

[1]CBA/N mice have an X-linked immunodeficiency gene (Xid) producing a defect in the Bruton tyrosine kinase (btk) associated with poor B-1 cell maturation and inadequate responses to type II T-independent antigens.
[2]Motheaten mice have the me^v mutation affecting the protein tyrosine phosphatase 1C (*PTP-1C*) gene which dramatically alters the threshold for antigen and strongly biases development toward the B-1 subpopulation. The mice have widespread autoimmunity and most of their B-cells are B-1.

B-1 AND B-2 CELLS REPRESENT TWO DISTINCT POPULATIONS

We have previously drawn attention to the subpopulation of B-cells which, in addition to surface IgM, express CD5 (cf. p. 208). The progenitors of this subset move from the fetal liver to the peritoneal cavity fairly early in life, at which stage they are the most abundant B-cell type and predominate in their contribution to the idiotype network and to the production of low affinity, multispecific IgM autoantibodies and the so-called 'natural' antibodies to bacterial carbohydrates, which seemingly arise slightly later in the neonatal period without obvious exposure to conventional antigens.

The **B-1 phenotype, viz. high surface IgM, low surface IgD, $CD43^+$ and $CD23^-$**, is shared by a minority subpopulation which is, however, $CD5^-$; these two populations are referred to as B-1a and B-1b respectively (figure 12.13). The **phenotype** of conventional **B-2 cells** (table 12.3), **low surface IgM, high surface IgD, $CD5^-$, $CD43^-$ and $CD23^+$**, reflects the fact that they represent a separate developmental lineage (figure 12.13). Some general comments may be in order. Although B-1 cells can shift to a B-2 phenotype, and possibly vice versa, there is minimal conversion between the two lineages under normal circumstances. The B-1 cells are particularly prevalent in the peritoneal cavity, main-

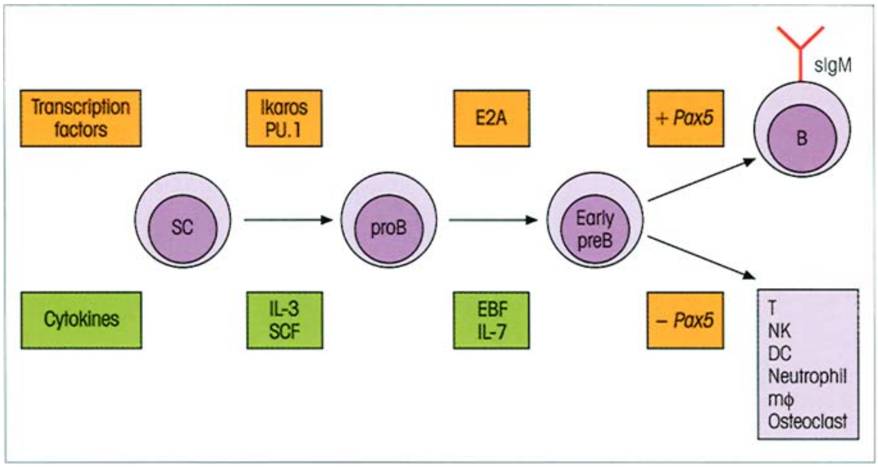

Figure 12.12. ***Pax5*** **is required for B-cell differentiation.** Hematopoietic stem cells (SC) under the influence of stem cell factor (SCF), IL-3 and the Ikaros and PU.1 transcription factors can differentiate into pro-B-cells. Further differentiation into pre-B-cells requires the E2A transcription factor together with early B-cell factor (EBF) and IL-7. Homozygous E2A mutant mice lack pre-B-cells, there being a block to D_HJ_H rearrangement in the Ig heavy chain locus plus severe reduction in RAG-1, Ig-α, CD19 and λ_5 transcripts. If at the early pre-B stage *Pax5* is not expressed, then differentiation along the B-cell lineage pathway comes to an abrupt halt. These early pre-B-cells have rearranged Ig D_H to J_H indicating their intention to become B-cells. However, even at this late stage, they can make other lineage choices as evidenced by the fact that, in the absence of *Pax5* expression, they can give rise to a number of other cell types if they are provided with appropriate cytokines. Indeed, *Pax5*$^{-/-}$ clones are able to develop into T-cells if transferred to immunodeficient mice, in which case they express rearranged TCR genes in addition to their initial Ig heavy chain gene rearrangement.

tain their numbers by self-replenishment and limit their *de novo* production from progenitors by feedback regulation. They can express both CD5 and its ligand CD72 on their surface, which should encourage mutual interaction, but a major factor influencing self-renewal could be the constitutive production of IL-10, since treatment of mice with anti-IL-10 from birth virtually wipes out the B-1 subset. The predisposition for self-renewal may underlie their undue susceptibility to become leukemic, with the malignant cells in chronic lymphocytic leukemia being almost invariably CD5$^+$.

B-1 cells tend to use particular germ-line *V* genes, and the autoantibody response to bromelain-treated erythrocytes is restricted to this subset which utilizes the rather diminutive V_H11 and V_H12 families. Clonal expansion seems to be driven by reaction with self-antigens (see legend to figure 12.13). They tend to respond to type 2 thymus-independent antigens (cf. p. 171) and, unlike the B-2 population, they do not enter into liaisons with thymus-dependent antigens, do not enter germinal centers and hence do not undergo somatic mutation or form high affinity antibodies. This may be just as well if the harmless low affinity autoantibodies which are produced by many B-1 cells are not automatically driven to high affinity pathogenic autoantibodies. In a weak moment one sometimes hears of 'good' and 'bad' autoantibodies, with the 'good guys' possibly having the job of sweeping up broken down self-components, as envisaged by dear Pierre Grabar many years ago when he thought of them as *globulines transporteurs*.

Other functions of B-1 cells may be the generation of an idiotype network concerned in self-tolerance, the response to conserved microbial antigens, and possibly the idiotypic regulation of B-2 responses. They are certainly the source of 'natural antibodies' which provide a pre-existing first line of IgM defense against common microbes. Up to 50% of the IgA-producing cells in the lamina propria are derived from peritoneal cavity B-1 cells. These cells are therefore an important source of the mucosal IgA which coats the normal microflora of the gut.

DEVELOPMENT OF B-CELL SPECIFICITY

The sequence of immunoglobulin gene rearrangements

By analysis of A-MuLV-transformed clones of pre-B-cells, it has proved possible to unravel the orderly cascade of Ig gene rearrangements which occur during differentiation.

Stage 1. Initially, the *D–J* segments on both heavy chain coding regions (one from each parent) rearrange (figure 12.14).

Stage 2. A *V–DJ* recombinational event now occurs on one heavy chain. If this proves to be a *nonproductive* re-

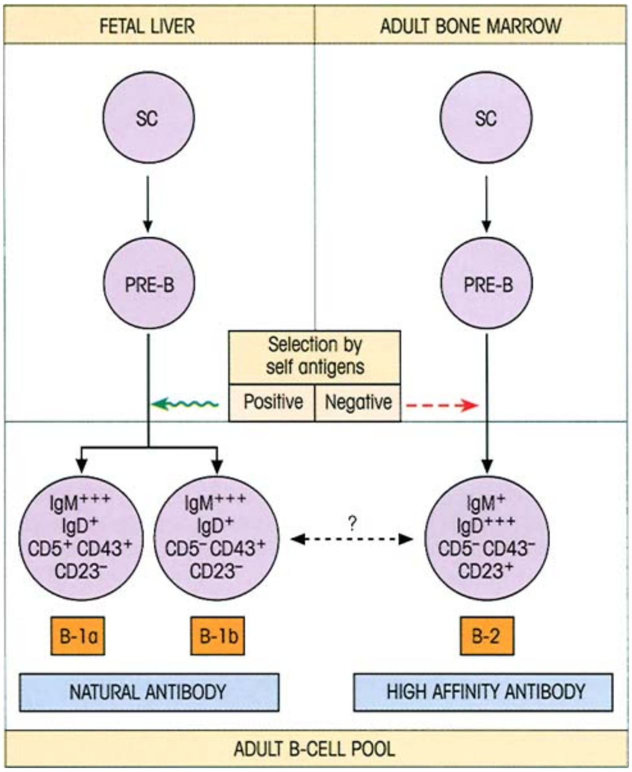

Figure 12.13. The development of separate B-cell subpopulations. It is presumed that B-1 cells of sufficiently high avidity for, say, self-surface antigens are eliminated, leaving positively selected lower affinity specificities for soluble self-antigens and the spectrum of 'natural antibody'-producing B-1 cells. In contrast, B-2 cells undergo negative, rather than positive, selection by self-antigens and the surviving B-2 cells give rise to the higher affinity IgG antibody produced by helper T-cell-dependent class-switched B-cells. It is thought that, although these subsets might be able to give rise to each other under some circumstances, generally they are maintained as separate lineages. Direct evidence that self-antigens positively select B-1 cells is provided by mice made transgenic for the V_H*3609* heavy chain gene. The transgene-encoded heavy chain pairs with endogenous Vκ21C light chain to produce an anti-thymocyte autoantibody associated with $CD5^+$ B-cells and which recognizes a Thy-1-associated carbohydrate epitope. High levels of the transgenic B-cells and of the serum autoantibody were found only in the presence of the autoantigen, being absent in Thy-1 knockout mice. SC, stem cell; CD23, FcεRII; CD43, leukosialin.

arrangement (i.e. adjacent segments are joined in an incorrect reading frame or in such a way as to generate a termination codon downstream from the splice point), then a second *V–DJ* rearrangement will occur on the sister heavy chain region. If a productive rearrangement is not achieved, we can wave the pre-B-cell a fond farewell.

Stage 3. Assuming that a productive rearrangement is made, the pre-B-cell can now synthesize μ chains. At around the same time, two genes, V_{preB} (*CD179a*) and λ_5 (*CD179b*), with homology for the V_L and C_L segments of λ light chains respectively, are temporarily transcribed to form a 'pseudo-light chain' which associates with the μ chains to generate a surface surrogate 'IgM' receptor, together with the Ig-α (CD79a) and Ig-β (CD79b) chains conventionally required to form a functional B-cell receptor. Expression of this receptor is absolutely essential for further differentiation of the B-lymphocytes since disruption of the membrane exon of the μ chain or of the λ_5 gene by homologous recombination of embryonic stem cells (cf. p. 141) arrests development at the pre-B stage and the animal is devoid of mature B-cells. This surrogate receptor closely parallels the pre-Tα/β receptor on pre-T-cell precursors of αβ TCR-bearing cells.

Stage 4. The surface receptor is signaled, perhaps by a stromal cell, to suppress any further rearrangement of heavy chain genes on a sister chromatid. This is termed **allelic exclusion** and was first discussed in relation to the rearrangement of TCR β chains (see p. 227).

Stage 5. It is presumed that the surface receptor now initiates the next set of gene rearrangements which occur on the κ light chain gene loci. These involve *V–J* recombinations on first one and then the other κ allele until a productive V_κ–*J* rearrangement is accomplished. Were that to fail, an attempt would be made to achieve productive rearrangement of the λ alleles. Synthesis of conventional sIgM now proceeds.

Stage 6. The sIgM molecule now prohibits any further gene shuffling by allelic exclusion of any unrearranged light chain genes.

At the next stage of differentiation, the cell develops a commitment to producing a particular antibody class and either bears surface IgM alone or in combination with IgA or IgG. The further addition of surface IgD now marks the readiness of the virgin B-cell for priming by antigen. Some cells, therefore, bear surface Ig of three different classes: M, G and D or M, A and D; but all Ig molecules on a single cell have the same idiotype and therefore are derived from the same V_H and V_L genes, presumably by splicing of a long RNA transcript. IgD is lost on antigenic stimulation so that memory cells lack this Ig. At the terminal stages in the life of a fully mature plasma cell, virtually all surface Ig is shed. Injection of anti-μ (anti-IgM heavy chain) into chick embryos prevents the subsequent maturation of IgM and IgG antibody-producing cells, whereas anti-γ inhibits only IgG development. Although we have seen earlier that T-helpers can induce class switching, it is also the case that some isotype switching probably occurs independently of antigen as a result of microenvironmental factors. In the embryonic chicken bursa, a regular switch from IgM to IgG is observed and it seems possible that local influences in the gut will prove to be responsible for the predominant

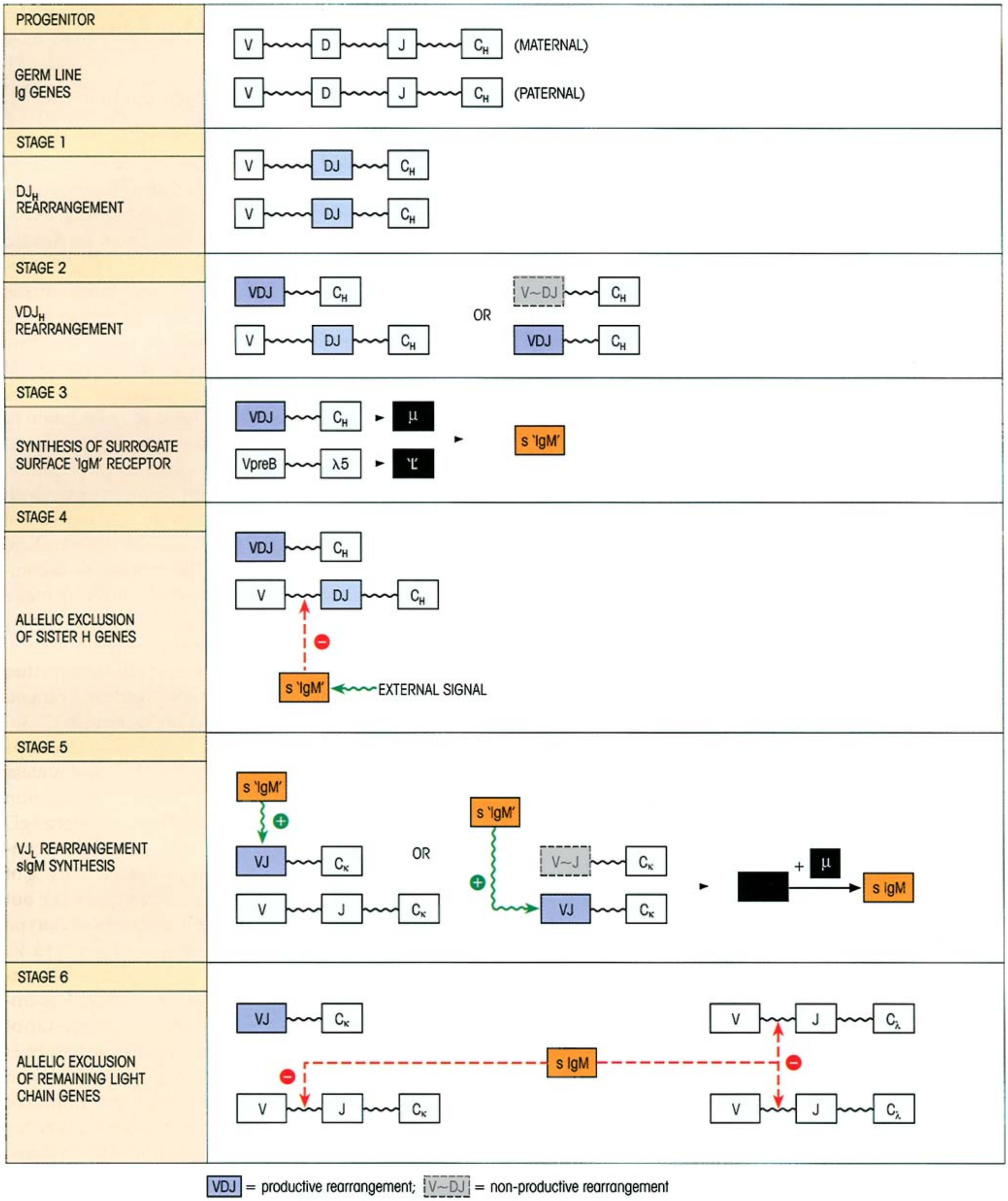

Figure 12.14. Sequence of B-cell gene rearrangements and postulated mechanism of allelic exclusion (see text).

development of IgA-bearing cells. These cells are generated in Peyer's patches, pass into the blood via the thoracic duct and return to populate the diffuse lymphoid tissue in the lamina propria of the gut.

The importance of allelic exclusion

Since each cell has chromosome complements derived from each parent, the differentiating B-cell has four light and two heavy chain gene clusters to choose from. We have described how, once the *VDJ* DNA rearrangement has occurred within one light and one heavy chain cluster, the *V* genes on the other four chromosomes are held in the embryonic state by an allelic exclusion mechanism so that the cell is able to express only one light and one heavy chain. This is essential for clonal selection to work since the cell is then only programed to make the one antibody it uses as a cell surface receptor to recognize antigen. Furthermore, this gene exclusion mechanism prevents the formation of molecules containing two different light or two different heavy chains which would have nonidentical combining sites and therefore be functionally monovalent with respect to the majority of antigens; such antibodies would be nonagglutinating and would tend to have low avidity as the bonus effect of multivalency could not operate.

Different specific responses can appear sequentially

The responses to given antigens in the neonatal period appear sequentially, as though each species were programed to rearrange its *V* genes in a definite order (figure 12.15). Early in ontogeny there is a bias favoring the rearrangement of the V_H genes most proximal to the *DJ* segment.

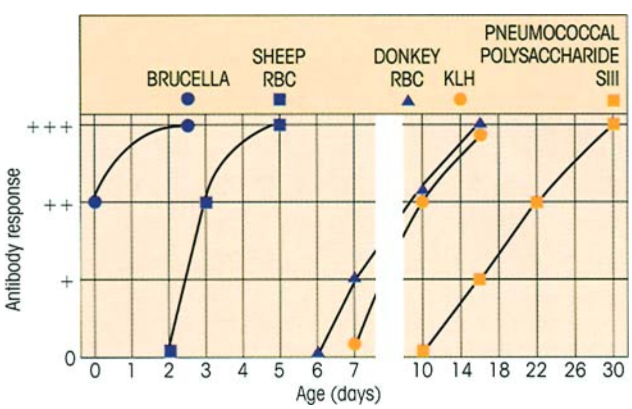

Figure 12.15. Sequential appearance of responsiveness to different antigens in the neonatal rat. RBC, red blood cell; KLH, keyhole limpet hemocyanin.

THE INDUCTION OF TOLERANCE IN B-LYMPHOCYTES

Tolerance can be caused by clonal deletion and clonal anergy

Just as for T-cells, so both mechanisms can operate on B-cells to prevent the reaction to self. Excellent evidence for deletion comes from mice bearing transgenes coding for IgM which binds to H-2K molecules of all *H-2* haplotypes except *d* and *f*. Mice of $H\text{-}2^a$ haplotype express the transgenic IgM abundantly in the serum, while 25–50% of total B-cells bear the transgenic idiotype. ($d \times k$) F1 crosses completely failed to express the transgene, either in the serum or on B-cells, i.e. B-cells programed for anti-H-$2K^k$ were expressed in $H\text{-}2^d$ mice but deleted in mice positive for H-$2K^k$ which in these circumstances acts as an autoantigen.

Tolerance through B-cell anergy was clearly demonstrated in another study in which double transgenic mice were made to express both lysozyme and a high affinity antibody to lysozyme. The animals were completely tolerant and could not be immunized to make anti-lysozyme; nor did the transgenic antibody appear in the serum although it was abundantly present on the surface of B-cells. These anergic cells could bind antigen to their surface receptors but could not be activated. Like the aged roué, wistfully drinking in the visual attractions of some young belle, these tolerized lymphocytes can 'see' the antigen but lack the ability to do anything about it.

Whether deletion or anergy is the outcome of the encounter with self probably depends upon the concentration and ability to cross-link Ig receptors. In the first of the two B-cell tolerance models above, the H-$2K^k$ autoantigen would be richly expressed on cells in contact with the developing B-lymphocytes and could effectively cause cross-linking. In the second case, the lysozyme, masquerading as a 'self'-molecule, is essentially univalent with respect to the receptors on an anti-lysozyme B-cell and would not readily bring about cross-linking. The hypothesis was tested by stitching a transmembrane hydrophobic segment onto the lysozyme transgene so that the antigen would be inserted into the cell membrane. Result? B-cells expressing the high affinity anti-lysozyme transgene were eliminated.

Another self-censoring mechanism, termed receptor editing, may come into play. We have already discussed one type of receptor editing (cf. p. 66) in which secondary rearrangements substitute another *V* gene onto an already rearranged *V*(*D*)*J* segment. However, receptor editing can also occur by wholesale replacement of an entire light chain. This can best be explained by an example. If the heavy and light chain Ig genes encoding a high affinity anti-DNA autoantibody are introduced as transgenes into a mouse, a variety of light chains are produced by genetic reshuffling until a combination with the heavy chain is achieved which no longer has anti-DNA activity, i.e. the autoreactivity is edited out. This will often involve replacement of a κ light chain with a new rearrangement made on the λ light chain locus and is associated with re-expression of the *RAG-1/2* genes.

Most peripheral B-cells in mice are ligand selected as revealed by analysis of the V_H repertoire at the cDNA level of bone marrow pre-B-cells compared with mature spleen B-cells. Once peripheralized, the bulk of the B-cell pool is stable; lymph node B- (and T-) cells from unprimed mice survived comfortably for at least 20 months on transfer to H-2 identical SCID animals.

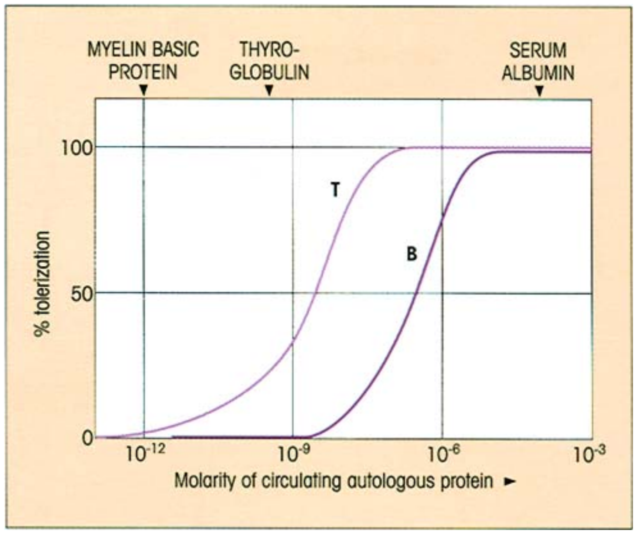

Figure 12.16. Relative susceptibility of T- and B-cells to tolerance by circulating self-antigens. Those circulating at low concentration induce no tolerance; at intermediate concentration, e.g. thyroglobulin, T-cells are moderately tolerized; molecules such as albumin which circulate at high concentrations tolerize both B- and T-cells.

Tolerance may result from helpless B-cells

With soluble proteins at least, T-cells are more readily tolerized than B-cells (figure 12.16) and, depending upon the circulating protein concentration, a number of self-reacting B-cells may be present in the body which cannot be triggered by T-dependent self-components since the T-cells required to provide the necessary T–B help are already tolerant—you might describe the B-cells as helpless. If we think of the determinant on a self-component which combines with the receptors on a self-reacting B-cell as a hapten and another determinant which has to be recognized by a T-cell as a carrier (cf. figure 9.11), then tolerance in the T-cell to the carrier will prevent the provision of T-cell help and the B-cell will be unresponsive. Take C5 as an example; this is normally circulating at concentrations which tolerize T- but not B-cells. Some strains of mice are congenitally deficient in C5 and their T-cells can help C5-positive strains to make antibodies to C5, i.e. the C5-positive strains still have inducible B-cells but they are helpless and need nontolerized T-cells from the C5-negative strain (figure 12.17).

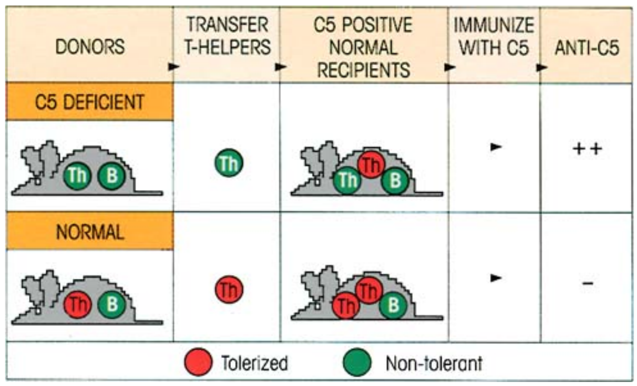

Figure 12.17. Circulating C5 tolerizes T- but not B-cells leaving them helpless. Animals with congenital C5 deficiency do not tolerize their T-helpers and can be used to break tolerance in normal mice.

It is worth noting the observation that injection of high doses of a soluble antigen without adjuvant, even when given several days after primary immunization with that antigen, prevented the emergence of high affinity mutated antibodies. Transfer experiments showed the T-cells to be tolerant. This tells us that, even when an immune response is well underway, T-helpers in the germinal center are needed to permit the mutations which lead to affinity maturation of antibody and, as a further corollary, that soluble self-antigens in the extracellular fluids can act to switch off autoreactive B-cells arising in the germinal centers by hypermutation.

Presumably, self-tolerance in both B- and T-cells involves all the mechanisms we have discussed to varying degrees and these are summarized in figure 12.18. Remember that, throughout the life of an animal, new stem cells are continually differentiating into immunocompetent lymphocytes and what is early in ontogeny for them can be late for the host; this means that self-tolerance mechanisms are still acting on early lymphocytes even in the adult, although it is always comforting to note that the threshold concentration for tolerance induction is very much lower for pre-B-cells relative to mature B-cells.

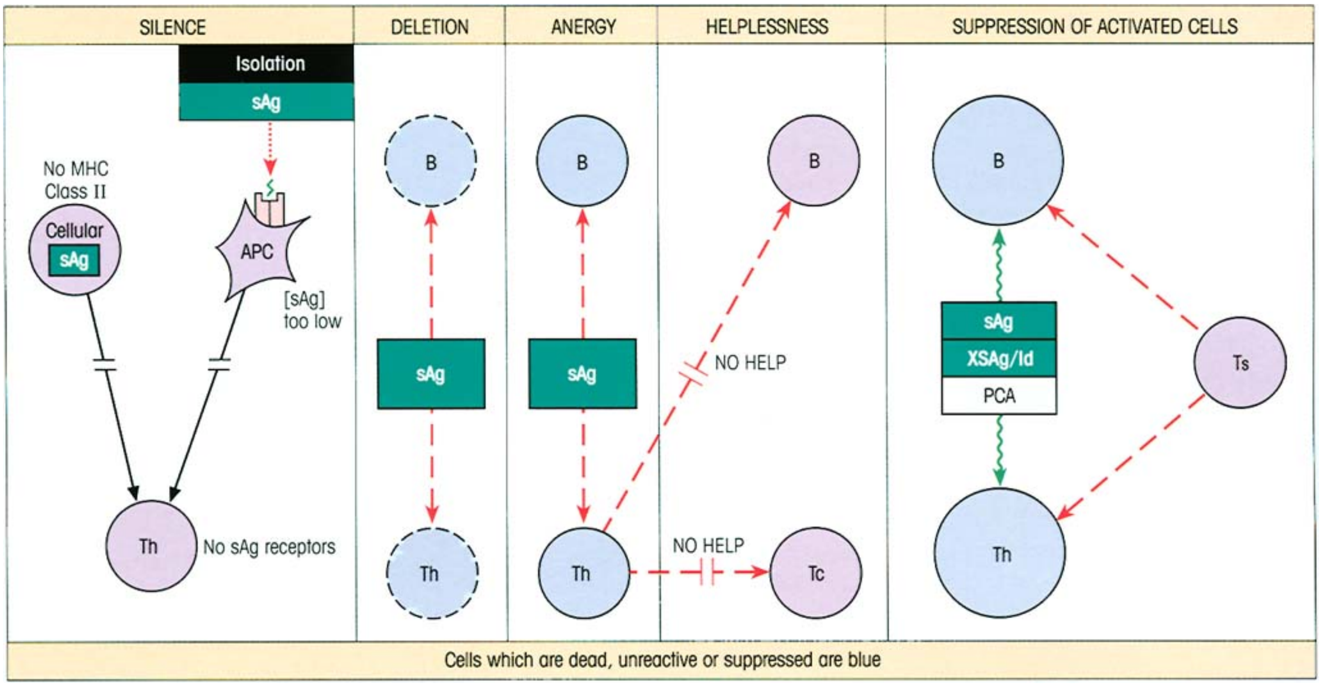

Figure 12.18. Mechanisms of self-tolerance (see text). sAg, self-antigen; XSAg/Id, cross-reacting antigen or idiotype; PCA, polyclonal activator; APC, antigen-presenting cell; Th, T-helper; Ts, T-suppressor; Tc, cytotoxic T-cell precursor.

NATURAL KILLER (NK) CELL ONTOGENY

The precise lineage of NK cells is still to be established. They share a common early progenitor with T-cells and express the CD2 molecule which is also present on T-cells. Furthermore, they have IL-2 receptors, are driven to proliferate by IL-2 and produce IFNγ. The majority of NK cells express CD56, a molecule also present on a subset of $CD4^+$ and $CD8^+$ T-cells. However, they do not develop in the thymus, their T-cell receptor *V* genes are not rearranged and, unlike T-cells, they express the CD16 FcγRIII. The current view is therefore that they separate from the T-cell lineage very early on in their differentiation.

THE OVERALL RESPONSE IN THE NEONATE

Lymph node and spleen remain relatively underdeveloped in the human at the time of birth, except where there has been intrauterine exposure to antigens as in congenital infections with rubella or other organisms. The ability to reject grafts and to mount an antibody response is reasonably well developed by birth, but the immunoglobulin levels, with one exception, are low, particularly in the absence of intrauterine infection. The exception is IgG which is acquired by placental transfer from the mother, a process dependent upon Fc structures specific to this Ig class. This material is catabolized with a half-life of approximately 30 days and there is a fall in IgG concentration over the first 3 months accentuated by the increase in blood volume of the growing infant. Thereafter, the rate of synthesis overtakes the rate of breakdown of maternal IgG and the overall concentration increases steadily. The other immunoglobulins do not cross the placenta and the low but significant levels of IgM in cord blood are synthesized by the baby (figure 12.19). IgM reaches adult levels by 9 months of age. Only trace levels of IgA, IgD and IgE are present in the circulation of the newborn.

THE EVOLUTION OF THE IMMUNE RESPONSE

Invertebrates have microbial defense mechanisms

Mechanisms for the recognition and subsequent **rejection of nonself** can be identified in invertebrates as far down the evolutionary scale as marine sponges, commonly regarded as the most primitive of present-day animals (figure 12.20). Phagocytosis is of importance throughout the animal kingdom (cf. Milestone 1.1, Chapter 1). In many phyla, phagocytosis is augmented by coating with agglutinins and bactericidins capable of binding to *p*athogen-*a*ssociated *m*olecular *p*atterns (PAMPs) on the microbial surface so providing the basis for the recognition of 'nonself'. It is notable that **infection very rapidly induces the synthesis of an impressive battery of antimicrobial peptides** in higher insects following activation of transcription factors

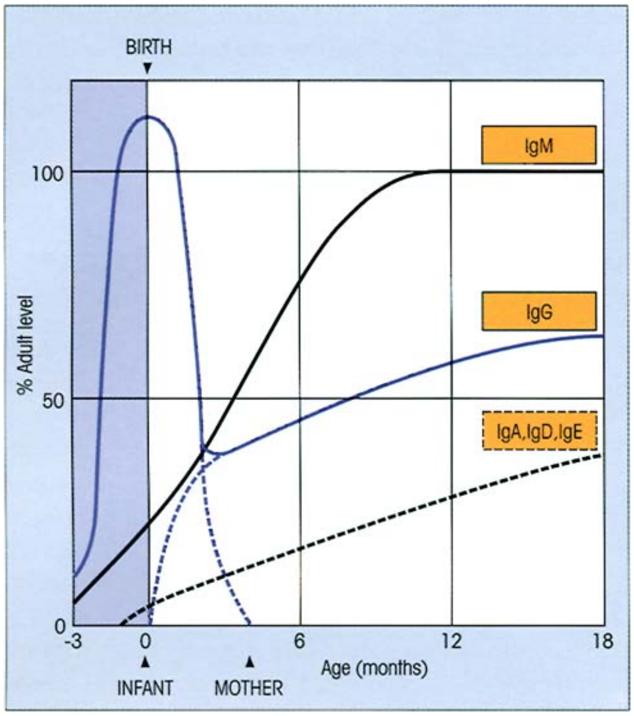

Figure 12.19. Development of serum immunoglobulin levels in the human. (After Hobbs J.R. (1969) In Adinolfi M. (ed.) *Immunology and Development*, p. 118. Heinemann, London.)

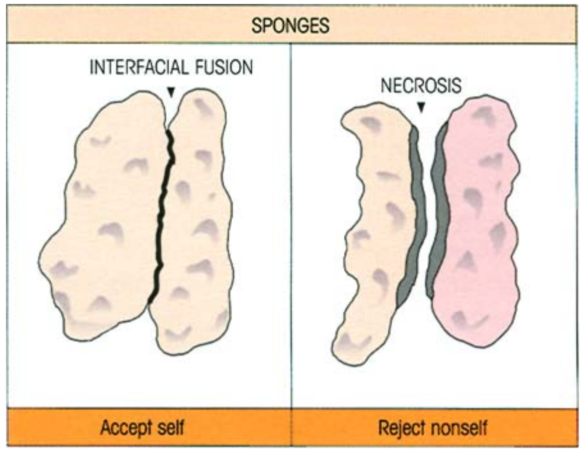

Figure 12.20. Recognition and rejection of nonself. Parabiosed fingers of a marine sponge from the same colony are permanently united but members of different colonies reject each other by 7–9 days.

which bind to promoter sequence motifs homologous to regulatory elements involved in the mammalian acute phase response. Thus, the toll molecule in drosophila is a receptor for PAMPs that activates NFκB in these flies. Drosophila with a loss-of-function mutation in *toll* are susceptible to fungal infections. Antimicrobial peptides produced by insects include disulfide-bridged cyclic peptides such as the 4 kDa anti-Gram-positive defensins and the 5 kDa antifungal peptide, drosomycin. Linear peptides inducible by infection include the cecropins and a series of anti-Gram-negative glycine- or proline-rich polypeptides. Cecropins, which have also been identified in mammals, are 4 kDa strongly cationic amphipathic α-helices causing lethal disintegration of bacterial membranes by creating ion channels.

Elements of a primordial complement system also exist among the lower orders. A protease inhibitor, an α_2-macroglobulin structurally homologous to C3 with internal thiolester, is present in the horseshoe crab. Conceivably this might represent an ancestral version of C3 which is activated by proteases released at a site of infection, deposited onto the microbe and recognized there as a ligand for the phagocytic cells. The complement receptor CR3 is an integrin, and related integrins in insects may harbor common ancestors. Mention of the horseshoe crab may have stirred a neuronal network in readers with good memories, to recall its synthesis of limulin (cf. p. 16) which is homologous with the mammalian acute phase C-reactive protein (CRP); presumably, it acts as a lectin to opsonize bacteria and is likely to be a product of the evolutionary line leading ultimately to C1q, mannose-binding protein and lung surfactant.

The other major strategy effectively deployed by invertebrates is to wall off an invading microorganism. This is achieved, for example, through proteolytic cascades which produce a coagulum of 'gelled' hemolymph around the offender.

Please do not let us overlook plants. Higher plants develop an 'immune state' of **systemic acquired resistance** (SAR), which can be established following a localized infection with pathogens that induce lesions involving host cell death. SAR persists for several weeks and extends to a broad range of bacterial, viral and fungal pathogens beyond the initiating infective agent. A series of SAR genes encode a wide variety of microbicidal proteins which can be induced through endogenous chemical mediators such as salicylic acid and methyl-2,6-dichloroisonicotinic acid. One function of salicylic acid is to bind to a protein with catalase activity, thereby increasing H_2O_2, but, while this may contribute to an acute defensive response, other mechanisms are thought to be concerned in the induction of SAR.

Adaptive immune responses appear with the vertebrates

Lower vertebrates

Lymphocytes and genuine adaptive T- and B-responses do not emerge in the phylogenetic tree until we reach the vertebrates, although neither can be elicited in the lowliest vertebrate studied—the California hagfish. This unpleasant cyclostome (which preys upon moribund fish by entering their mouths and eating the flesh from the inside) can respond to hemocyanin provided that it is maintained at temperatures approaching 20°C (in general, poikilotherms make antibodies better at higher temperatures), but true immunoglobulins are not involved. Further up the evolutionary scale in the cartilaginous fishes, well-defined 18*S* and 7*S* immunoglobulins with heavy and light chains have now been defined, but the responses are *T-independent*.

T-cells appear

The toad, *Xenopus*, is a pliable, if unlovely, species for study since it is possible to make transgenics and cloned tadpoles fairly readily, and it has a less complex lymphoid system than mammals, characterized by a small number of lymphocytes and a restricted antibody repertoire not subject to somatic mutation. Furthermore, positive and negative thymic selection have been demonstrated in frogs.

The emergence of an honest-to-God thymus in the teleosts (bony fishes), amphibians, reptiles, birds and mammals was of course associated with MHC molecules, cell-mediated immunity, cytotoxic T-cells and allograft rejection. It could be argued that we also see phylogenetically more ancient, T-independent B-1 (CD5-positive) cells joined by a new T-dependent B-2 population. However, T-dependent, high affinity, heterogeneous, rapid secondary antibody responses are only seen with warm-blooded vertebrates such as birds and mammals, and these correlate directly with the evolution of germinal centers.

Generation of antibody diversity

Mechanisms for the generation of antibody diversity receive quite different emphasis as one goes from one species to another. We are already familiar with the mammalian system where multiple *V* genes are greatly amplified by a variety of recombinational events involving multiple *D* and *J* segments. The horned shark also has many *V* genes, but the opportunities for combinatorial joining are tightly constrained by close linkage between individual *V*, *D*, *J* and *C* segments and this may be a factor in the restricted antibody response of this species. In sharp contrast, there seems to be only one operational *V* gene at the light chain locus in the chicken, but this undergoes extensive somatic diversification utilizing nonfunctional adjoining *V* pseudogenes in a somatic gene conversion-like process. Camel lovers should note that not only do they get by on little water but, like the llamas, they also survive on antibodies which lack light chains.

THE EVOLUTION OF DISTINCT B- AND T-CELL LINEAGES WAS ACCOMPANIED BY THE DEVELOPMENT OF SEPARATE SITES FOR DIFFERENTIATION

The differential effects of neonatal bursectomy and thymectomy in the chicken on subsequent humoral and cellular responses paved the way for our eventual recognition of the separate lymphocyte lineages which subserve these functions. Like the thymus, the bursa of Fabricius develops as an embryonic outpushing of the gut endoderm, this time from hindgut as distinct from foregut, and provides the microenvironment to cradle incoming stem cells and direct their differentiation to immunocompetent B-lymphocytes. As may be seen from table 12.4, neonatal bursectomy had a profound effect on overall immunoglobulin levels and on specific antibody production following immunization, but did not unduly influence the cell-mediated delayed-type hypersensitivity (DTH) response to tuberculin or affect graft rejection. On the other hand, thymectomy grossly impaired cell-mediated reactions and inhibited antibody production to most protein antigens.

The distinctive anatomical location of the B-cell differentiation site in a separate lymphoid organ in the chicken was immensely valuable to progress in this field because it allowed the above types of experiments to be carried out. However, many years went by in a fruitless search for an equivalent bursa in mammals before it was realized that the primary site for B-cell generation was in fact the bone marrow itself.

Table 12.4. Effect of neonatal bursectomy and thymectomy on the development of immunologic competence in the chicken. (From Cooper M.D., Peterson R.D.A., South M.A. & Good R.A. (1966) *Journal of Experimental Medicine* **123**, 75, with permission of the editors.)

All X-irradiated after birth	Peripheral blood lymphocyte count	Ig conc.	Antibody	Delayed skin reaction to tuberculin	Graft rejection
Intact	14 800	++	+++	++	+++
Thymectomized	9 000	++	+	–	+
Bursectomized	13 200	–	–	+	+

CELLULAR RECOGNITION MOLECULES EXPLOIT THE IMMUNOGLOBULIN GENE SUPERFAMILY

When nature fortuitously chances upon a protein structure ('motif' is the buzz word) which successfully mediates some useful function, the selective forces of evolution make sure that it is widely exploited. Thus, all the molecules involved in antigen recognition which we have described at such (painful!) length in Chapters 3 and 4 are members of a gene superfamily related by sequence and presumably a common ancestry. All polypeptide members of this family, which includes heavy and light Ig chains, T-cell receptor α and β chains, MHC class I and class II molecules and β_2-microglobulin, are composed of one or more immunoglobulin homology units. Each unit is roughly 110 amino acids in length and is characterized by certain conserved residues around the two cysteines found in every domain and the alternating hydrophobic and hydrophilic amino acids which give rise to the familiar antiparallel β-pleated strands with interspersed short variable lengths having a marked propensity to form reversed turns—the '**immunoglobulin fold**' in short (cf. p. 45).

Attention has been drawn to a very important feature of the Ig domain structure, namely the mutual complementarity which allows strong interdomain noncovalent interactions, such as those between V_H and V_L and the two C_H3 regions which form the IgG pFc′ fragment. Gene duplication and diversification can create mutual families of interacting molecules, such as CD4 with MHC class II, CD8 with MHC class I and IgA with the poly-Ig receptor (figure 12.21). Likewise, the intercellular adhesion molecules ICAM-1 and N-CAM (figure 12.21) are richly endowed with these domains, and the long evolutionary history of N-CAM strongly suggests that these structures made an early appearance in phylogeny as mediators of intercellular recognition. In marine sponges, Ig superfamily structures are found both on the extracellular portion of the receptor tyrosine kinase (RTK) and in the more recently described cell recognition molecules (CRMs), both thought to be involved in allograft rejection. A recent trawl of the protein sequence database revealed hundreds of known members of the Ig superfamily. Some family!

The **integrins** form another structural superfamily which includes a number of hematopoietic cell surface molecules concerned with adhesion to extracellular matrix proteins and to cell surface ligands; their function is to direct leukocytes to particular tissue sites (see discussion on p. 223).

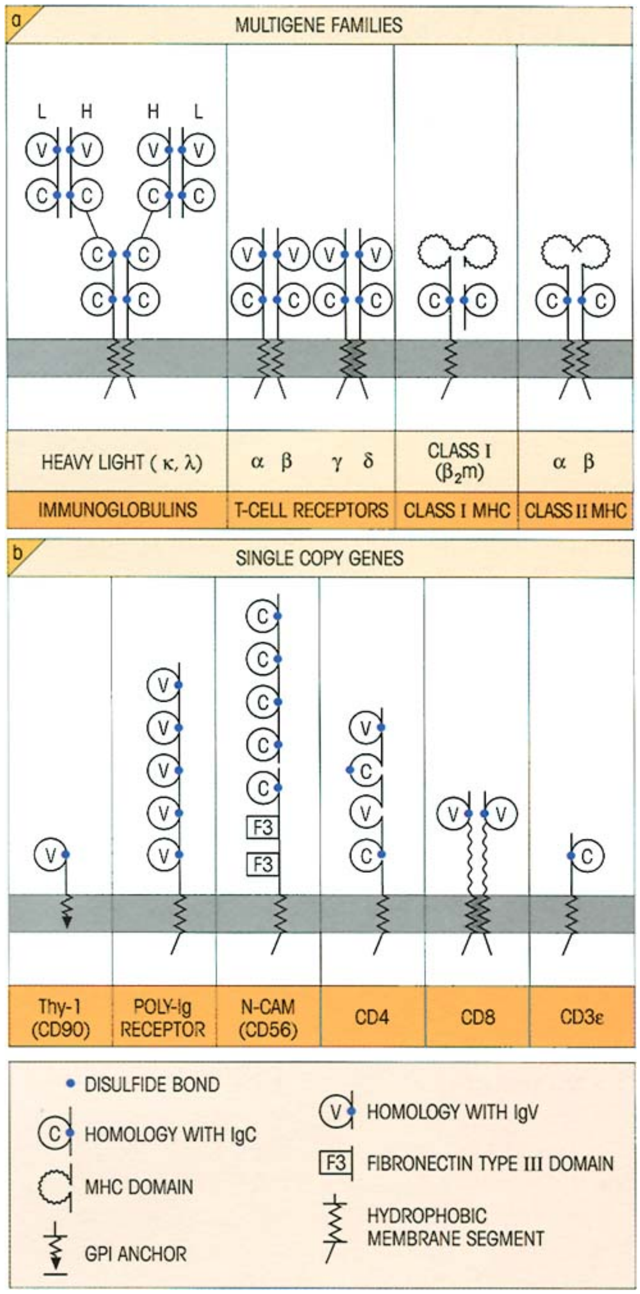

Figure 12.21. The immunoglobulin superfamily comprises a large number of surface molecules which all share a common structure, the immunoglobulin-type domain, suggesting evolution from a single primordial ancestral gene. Just a few examples are shown. (a) Multigene families involved in antigen recognition (the single copy β_2-microglobulin is included because of its association with class I). (b) Single copy genes. Thy-1 is present on T-cells and neurons. The poly-Ig receptor transports IgA across mucosal membranes. N-CAM is an adhesion molecule binding neuronal cells together. It is also found on NK cells and a subpopulation of T-cells, but its function on these lymphoid cells is unknown. (Reprinted by permission from *Nature* **323**, 15. Copyright © 1986, Macmillan Magazines Ltd with some updating.)

SUMMARY

Multipotential stem cells from the bone marrow give rise to all the formed elements of the blood

- Expansion and differentiation are driven by soluble growth (colony-stimulating) factors and contact with reticular stromal cells.

The differentiation of T-cells occurs within the microenvironment of the thymus

- Precursor T-cells arising from stem cells in the bone marrow need to travel to the thymus under the influence of chemokines in order to become immunocompetent T-cells.

T-cell ontogeny

- Differentiation to immunocompetent T-cell subsets is accompanied by changes in the surface phenotype which can be recognized with monoclonal antibodies.
- TCR genes rearrange in the thymus cortex, producing a $\gamma\delta$ TCR or a pre-$\alpha\beta$ TCR, consisting of an invariant pre-Tα associated with a conventional Vβ, before final rearrangement of the Vα to generate the mature $\alpha\beta$ TCR.
- Double-negative $CD4^-8^-$ pre-T-cells are driven and expanded, probably by Notch-mediated and other signals, to become double positive $CD4^+8^+$.
- The thymus epithelial cells **positively select** $CD4^+8^+$ T-cells with avidity for their MHC haplotype so that single-positive $CD4^+$ or $CD8^+$ T-cells develop that are restricted to the recognition of antigen in the context of the epithelial cell haplotype.

T-cell tolerance

- The induction of immunological tolerance is necessary to avoid self-reactivity.
- High avidity T-cells which react with self-antigens presented by corticomedullary macrophages and interdigitating dendritic cells are eliminated by **negative selection**. The paradigm that low avidity binding to MHC–peptide produces positive selection and high avidity negative, is probably broadly true but may need some amendment.
- Self-tolerance can also be achieved by anergy.
- Anergic cells attached to a dendritic cell can downregulate the antigen-presenting ability of that cell, resulting in infectious anergy.
- A state of what is effectively self-tolerance also arises when there is a failure to adequately present a self-antigen to lymphocytes, either because of compartmentalization, lack of class II on the antigen-presenting cell or low concentration of peptide–MHC (cryptic self).
- T-suppression is probably more concerned in reversing autoimmunity than preventing it.

B-cells differentiate in the fetal liver and then in the bone marrow

- They become immunocompetent B-cells after passing through pro-B-, pre-B- and immature B-cell stages.
- *Pax5* expression is essential for progression from the pre-B- to immature B-cell stage.

B-1 and B-2 represent two distinct subpopulations of B-cells

- B-1 cells represent a minor population expressing high sIgM and low sIgD. B-1a cells are $CD5^+$, B-1b are $CD5^-$. The majority of conventional B-cells, the B-2 population, are $sIgM^{lo}$, $sIgD^{hi}$, $CD5^-$. The B-1 population predominates in early life, shows a high level of idiotype–anti-idiotype connectivity, produces low affinity, IgM polyreactive antibodies, many of them autoantibodies, and is responsible for the T-independent 'natural' IgM antibacterial antibodies which appear spontaneously.

Development of B-cell specificity

- The sequence of Ig variable gene rearrangements is *DJ* and then *VDJ*.
- *VDJ* transcription produces μ chains which associate with $V_{preB} \cdot \lambda_5$ chains to form a surrogate surface IgM-like receptor.
- This receptor signals allelic exclusion of unrearranged heavy chains and initiates rearrangement of $V\text{–}J_\kappa$ (in the mouse) and, if unproductive, $V\text{–}J_\lambda$.
- If the rearrangement at any stage is unproductive, i.e. does not lead to an acceptable gene reading frame, the allele on the sister chromosome is rearranged.
- The mechanisms of allelic exclusion ensure that each lymphocyte is programed for only one antibody. Responses to different antigens appear sequentially with age.

The induction of tolerance in B-lymphocytes

- B-cell tolerance is induced by clonal deletion, clonal anergy, receptor editing and 'helplessness' due to preferential tolerization of T-cells needed to cooperate in B-cell stimulation.

Natural killer (NK) cell ontogeny

- NK cells share an early progenitor with T-cells but separate from the T-cell lineage early on and develop somewhere other than the thymus.

(*continued*)

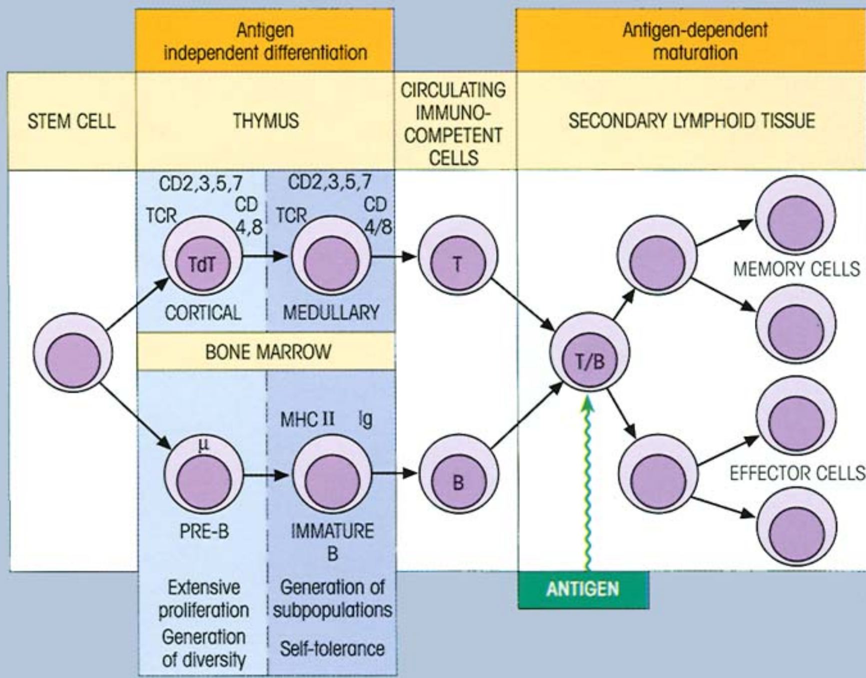

Figure 12.22. Antigen-independent differentiation and antigen-dependent maturation of T- and B-cells. Cortical thymocytes are positively selected to recognize self-MHC haplotype. TdT, terminal deoxynucleotidyl transferase.

The overall response in the neonate

- Maternal IgG crosses the placenta and provides a high level of passive immunity at birth.

The antigen-independent differentiation within primary lymphoid organs and antigen-driven maturation in secondary lymphoid organs are summarized in figure 12.22.

The evolution of the immune response

- Recognition of self is of fundamental importance for multicellular organisms, even lowly forms like marine sponges.
- Invertebrates have defense mechanisms based on phagocytosis, killing by a multiplicity of microbicidal peptides and polyphenoloxidase metabolites, and imprisonment of the invader by coagulation of the hemolymph.
- Higher plants can establish a persisting state of systemic acquired resistance.
- B- and T-cell responses are well defined in the vertebrates and the evolution of these distinct lineages was accompanied by the development of separate sites for differentiation.
- The success of the immunoglobulin domain structure, possibly through its ability to give noncovalent mutual binding, has been exploited by evolution to produce the very large Ig superfamily of recognition molecules, including Ig, TCRs, MHC class I and II, β_2-microglobulin, CD4, CD8, the poly-Ig receptor and Thy-1. Another superfamily, the integrins, which includes LFA-1 and the VLA molecules, is concerned with leukocyte binding to endothelial cells and extracellular matrix proteins.

See the accompanying website (**www.roitt.com**) for multiple choice questions.

FURTHER READING

von Boehmer H. (2000) T-cell lineage fate: instructed by receptor signals? *Current Biology* **10**, R642.

Goldrath A.W. & Bevan M.J. (1999) Selecting and maintaining a diverse T-cell repertoire. *Nature* **402**, 255.

Hardy R.R. & Hayakawa K. (2001) B cell development pathways. *Annual Review of Immunology* **19**, 595.

Herzenberg L.A. (2000) B-1 cells, the lineage question revisited. *Immunological Reviews* **175**, 9.

Horton J. & Ratcliffe N. (1998) Evolution of immunity. In Roitt I.M., Brostoff J. & Male D.K. (eds) *Immunology*, 5th edn, p. 199. Mosby, London.

Kamradt T. & Mitchison N.A. (2001) Tolerance and autoimmunity. *New England Journal of Medicine* **344**, 655.

Nutt S.L., Heavey B., Rolink A.G. & Busslinger M. (1999) Commitment to the B lymphoid lineage depends on the transcription factor Pax5. *Nature* **401**, 556.

Phillips R.L. *et al.* (2000) The genetic program of hematopoietic stem cells. *Science* **288**, 1635.

Robey E. & Fowlkes B.J. (1998) The αβ versus γδ T-cell lineage choice. *Current Opinion in Immunology* **10**, 181.

Townsend S.E., Weintraub B.C. & Goodnow C.C. (1999) Growing up on the streets: why B-cell development differs from T-cell development. *Immunology Today* **20**, 217.

Turner R.J. (ed.) (1994) *Immunology: A Comparative Approach*. John Wiley & Sons Ltd, Chichester.

Yasutomo K., Doyle C., Miele L. & Germain R.N. (2000) The duration of antigen receptor signalling determines $CD4^+$ versus $CD8^+$ T-cell lineage fate. *Nature* **404**, 506.

CHAPTER 13

Adversarial strategies during infection

Introduction, 249
Inflammation revisited, 249
Mediators of inflammation, 249
Leukocytes bind to endothelial cells through paired adhesion molecules, 250
Initiation of the acute inflammatory response, 250
The ongoing inflammatory process, 250
Regulation and resolution of inflammation, 252
Chronic inflammation, 253
Extracellular bacteria susceptible to killing by phagocytosis and complement, 254
Bacterial survival strategies, 254
The host counter-attack, 256
Some specific bacterial infections, 261
Bacteria which grow in an intracellular habitat, 262
Bacterial gambits, 262
Defense is by T-cell-mediated immunity (CMI), 263
Activated macrophages kill intracellular parasites, 264
Examples of intracellular bacterial infections, 264
Immunity to viral infection, 267
Immunity can be evaded by antigen changes, 267
Viruses can interfere with immune effector mechanisms, 268
Protection by serum antibody, 269
Local factors, 269
Cell-mediated immunity gets to the intracellular virus, 269
Immunity to fungi, 271
Immunity to parasitic infections, 271
The host responses, 272
Evasive strategies by the parasite, 275
Transmissible spongiform encephalopathies, 277
Immunopathology, 277
Summary, 277
Further reading, 280

INTRODUCTION

We are engaged in constant warfare with the microbes which surround us and the processes of mutation and evolution have tended to select microorganisms which have evolved means of evading our defense mechanisms. Pathogens continue to take a terrifying toll (figure 13.1), particularly in the developing world. Furthermore, the decline in mortality from infectious disease which had been seen in countries such as the USA has gone into reverse, with a rise in deaths in that country from infectious disease increasing at a rate of 4.8% per year between 1981 and 1995. There has been both the emergence of new infections and the re-emergence of some old adversaries. Among the current list of problems are Legionnaire's disease, HIV, ebola, nvCJD, *E. coli*, methicillin-resistant *Staphylococcus aureus* (MDSA), toxic shock syndrome and lyme disease. The fact that MDSA strains are becoming resistant to the drug of last resort, vancomycin, is, to put it mildly, worrying. Furthermore, it is becoming increasingly appreciated that infectious agents are related to many 'noninfectious' diseases, such as the association between *Helicobacter pylori* and peptic ulcer and of various viruses with cancer. In this chapter, we look at the varied, often ingenious, adversarial strategies which we and our enemies have developed over very long periods of time.

INFLAMMATION REVISITED

The acute inflammatory process involves a protective influx of white cells, complement, antibody and other plasma proteins into a site of infection or injury and was discussed in broad outline in the introductory chapters. Now that we are ready to look in more detail at the aggressive gambits and wily counter-attacks which characterize the conflict between microbe and host, it is appropriate to re-examine the mechanisms of inflammation in greater depth. The reader may find it helpful to have another look at the relevant sections in Chapters 1 and 2, particularly those relating to figures 1.15, 1.16, 1.17 and 2.18.

Mediators of inflammation

A complex variety of mediators are involved in acute inflammatory responses (figure 13.2). Some act directly on the smooth muscle wall surrounding the

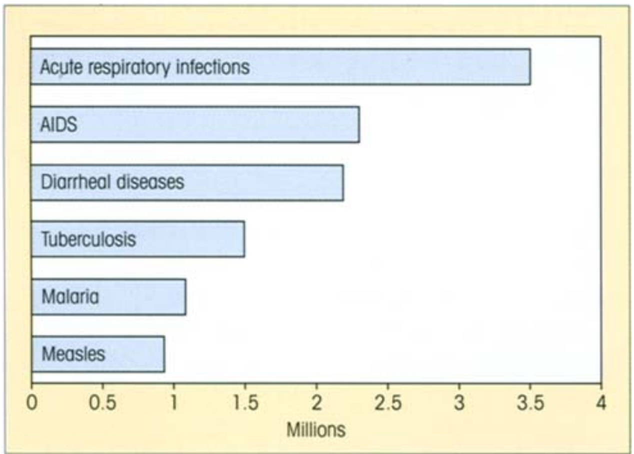

Figure 13.1. Deaths from infectious disease. These six diseases caused almost 90% of the 13 million deaths from infectious disease worldwide, as estimated by the World Health Organization for the year 1998. For more information, see http://www.who.int/infectious-disease-report.

arterioles to alter blood flow. Others act on the venules to cause contraction of the endothelial cells with transient opening of the interendothelial junctions and consequent transudation of plasma. The migration of leukocytes from the bloodstream is facilitated by mediators which upregulate the expression of adherence molecules on both endothelial and white cells and others which lead the leukocytes to the inflamed site through chemotaxis.

Leukocytes bind to endothelial cells through paired adhesion molecules

Redirecting the leukocytes charging along the blood into the site of inflammation is somewhat like having to encourage bulls stampeding down the Pamplona main street to move quietly into the side roads. We have had occasion to confront this problem when discussing lymphocyte homing (cf. p. 152, figure 8.6) and, in the present context, the adherence of leukocytes to the endothelial vessel wall through the interaction of complementary binding of cell surface molecules is an absolutely crucial step. Several classes of molecule subserve this function (cf. p. 152, table 8.3), some acting as lectins to bind a carbohydrate ligand on the complementary partner.

Initiation of the acute inflammatory response

A very early event is the upregulation of P-selectin and *p*latelet *a*ctivating *f*actor (PAF) on the endothelial cells lining the venules by histamine or thrombin released by the original inflammatory stimulus. Recruitment of these molecules from intracellular storage vesicles ensures that they appear within minutes on the cell surface. Engagement of the lectin-like domain at the tip of the P-selectin molecule with sialyl Lewis^x carbohydrate determinants on the mucin-like *P*-*s*electin *g*lycoprotein *l*igand-1 (PSGL-1) on the neutrophil surface causes the cell to slow and then **roll** along the endothelial wall and helps PAF to dock onto its corresponding receptor. This, in turn, increases surface expression of the integrins *l*ymphocyte *f*unction-*a*ssociated molecule-1 (LFA-1) and Mac-1 which now bind the neutrophil very firmly to the endothelial surface (figure 13.3).

Activation of the neutrophils also makes them more responsive to chemotactic agents and, under the influence of C5a and leukotriene-B4, they exit from the circulation by moving purposefully through the gap between endothelial cells, across the basement membrane (**diapedesis**) and up the chemotactic gradient to the inflammation site.

Damage to vascular endothelium, which exposes the basement membrane, and bacterial toxins, such as LPS, trigger other complex systems (figure 13.4). Activation of platelets by contact with basement membrane collagen or induced endothelial PAF leads to aggregation and **thrombus** formation by adherence through platelet glycoprotein Ib to von Willebrand factor on the vascular surface. Such platelet plugs are adept at stemming the loss of blood from a damaged artery, but in the venous system the damaged site is sealed by a **fibrin clot** resulting from activation of the intrinsic clotting system via contact of Hageman factor (factor XII) with the exposed surface of the basement membrane. Activated Hageman factor also triggers the kinin and plasmin systems and several of the resulting products influence the inflammatory process by increasing vascular permeability, activating endothelium, autocatalytically amplifying the production of Hageman factor XII and cleaving C3 (figure 13.4).

The ongoing inflammatory process

One must not ignore the role of the tissue macrophage which, under the stimulus of local infection or injury, secretes an imposing array of mediators. In particular, the cytokines interleukin-1 (IL-1) and tumor necrosis factor (TNF) act at a later time than histamine or thrombin to stimulate the endothelial cells and maintain the inflammatory process by upregulating E-selectin and sustaining P-selectin expression. Thus, expression of E-selectin occurs 2–4 hours after the initiation of acute inflammation, being dependent upon activation of

MEDIATOR ACTION				UPREGULATE ADHESION MOL.		
	DILATATION	CONSTRICTION	INCREASE PERMEABILITY	ENDOTHELIUM	PMN	PMN CHEMOTAXIS
HISTAMINE	+		+	++		
BRADYKININ	+		++			
PGE_2/I_2	+++		Potentiate other mediators			
VIP	+++					
NITRIC OXIDE	+++		+++			
LEUKOTRIENE-D4		+				
LEUKOTRIENE-C4		++	+			
C5a			++	+	++	+++
LEUKOTRIENE-B4			++		++	+++
f.Met.Leu.Phe			++		+	+
PLATELET ACTIVATING FACTOR	+		++		++	
IL-8					+++	+++
NAP-2 (CXCL7)					++	++
IL-1				++	++	
TNF				++	++	

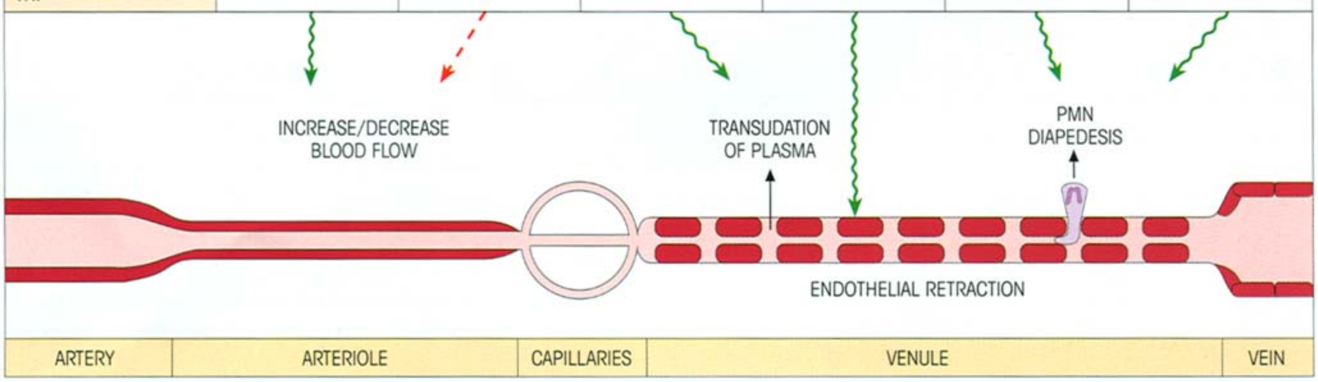

Figure 13.2. The principal mediators of acute inflammation. The reader should refer back to figure 1.15 to recall the range of products generated by the mast cell. The later acting cytokines such as interleukin-1 (IL-1) are largely macrophage-derived and these cells also secrete prostaglandin E_2 (PGE_2), leukotriene-B4 and the neutrophil activating chemokine NAP-2 (CXCL7). PMN, polymorphonuclear neutrophil; VIP, vasoactive intestinal peptide; TNF, tumor necrosis factor.

gene transcription. The E-selectin engages the glycoprotein E-selectin ligand-1 (ESL-1) on the neutrophil. Other later acting components are the **chemokines** (*chemo*tactic cyto*kines*) IL-8 (CXCL8) and epithelial-derived neutrophil attractant-78 (ENA-78, CXCL5) which are highly effective neutrophil chemoattractants. IL-1 and TNF also act on endothelial cells, fibroblasts and epithelial cells to stimulate secretion of another chemokine, MCP-1 (CCL2), a chemotactic protein for several different cell types which is particularly potent at attracting mononuclear phagocytes to the inflammatory site to strengthen and maintain the defensive reaction to infection.

Perhaps this is a good time to remind ourselves of the important role of chemokines (see table 10.3) in selectively attracting multiple types of leukocytes to inflammatory foci. Inflammatory chemokines are typically induced by microbial products such as lipopolysaccharide (LPS) and by proinflammatory cytokines including IL-1, TNF and IFNγ. As a very broad generalization, chemokines of the CXC subfamily, such as IL-8, are specific for neutrophils and, to varying extents, lymphocytes, whereas chemokines with the CC motif are chemotactic for T-cells, monocytes, dendritic cells, and variably for natural killer (NK) cells, basophils and eosinophils. Eotaxin (CCL11) is chemotactic for eosinophils, and the presence of significant concentrations of this mediator together

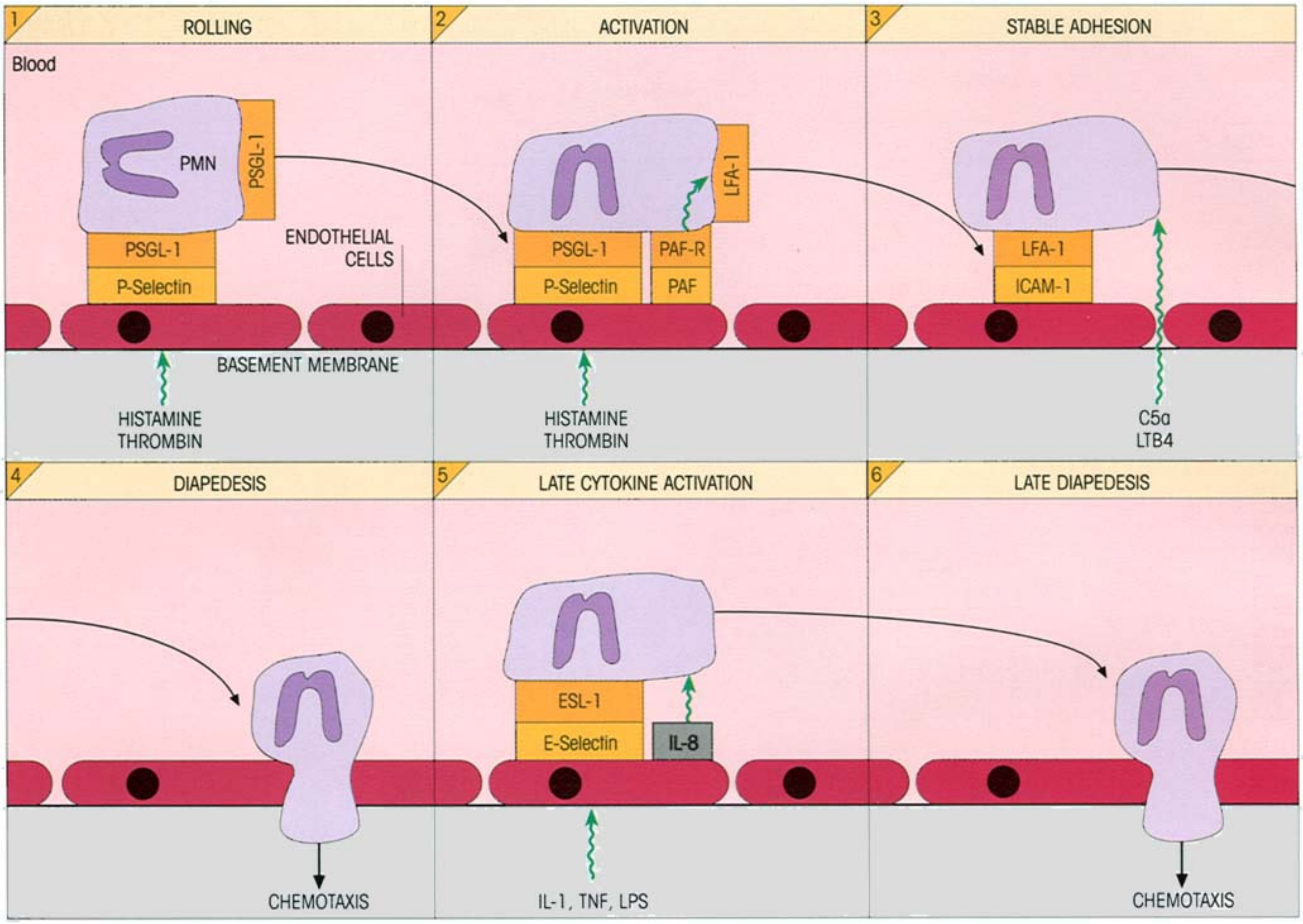

Figure 13.3. Early events in inflammation affecting neutrophil margination and diapedesis. Induced upregulation of P-selectin on the vessel walls plays the major role in the initial leukocyte–endothelial interaction (rolling) by interaction with ligands on the neutrophil such as the mucin-like P-selectin glycoprotein ligand-1 (PSGL-1, CD162). Recognition of extracellular gradients of the chemotactic mediators by receptors on the polymorphonuclear neutrophil (PMN) surface triggers intracellular signals which generate motion. The neutrophils crawl rather than swim and migration along the extracellular matrix vitronectin is dependent upon very rapid cycles of integrin-dependent adhesion and detachment regulated by calcineurin. The cytokine-induced expression of E-selectin, which is recognized by the glycoprotein E-selectin ligand-1 (ESL-1) on the neutrophil, occurs as a later event. Chemotactic factors such as IL-8, which is secreted by a number of cell types including the endothelium itself, are important mediators of the inflammatory process. (Compare events involved in homing and transmigration of lymphocytes, figure 8.6.)

with RANTES (*r*egulated upon *a*ctivation *n*ormal *T*-cell *e*xpressed and *s*ecreted, CCL5) in mucosal surfaces could account for the enhanced population of eosinophils in those tissues. The different chemokines bind to particular heparin and heparan sulfate glycosaminoglycans so that, after secretion, the chemotactic gradient can be maintained by attachment to the extracellular matrix as a form of scaffolding.

Clearly, this whole operation serves to focus the immune defenses around the invading microorganisms. These become coated with antibody, C3b and certain acute phase proteins and are ripe for phagocytosis by the granulocytes and macrophages; under the influence of the inflammatory mediators these have upregulated C3 and Ig receptors, enhanced phagocytic responses and hyped-up killing powers, adding up to bad news for the bugs.

Of course it is beneficial to recruit lymphocytes to sites of infection and we should remember that endothelial cells in these areas express VCAM-1 (cf. p. 151) which acts as a homing receptor for VLA-4-positive activated memory T-cells, while many chemokines are chemotactic for lymphocytes.

Regulation and resolution of inflammation

With its customary prudence, evolution has established regulatory mechanisms to prevent inflammation from getting out of hand. At the humoral level we have a series of complement regulatory proteins: C1

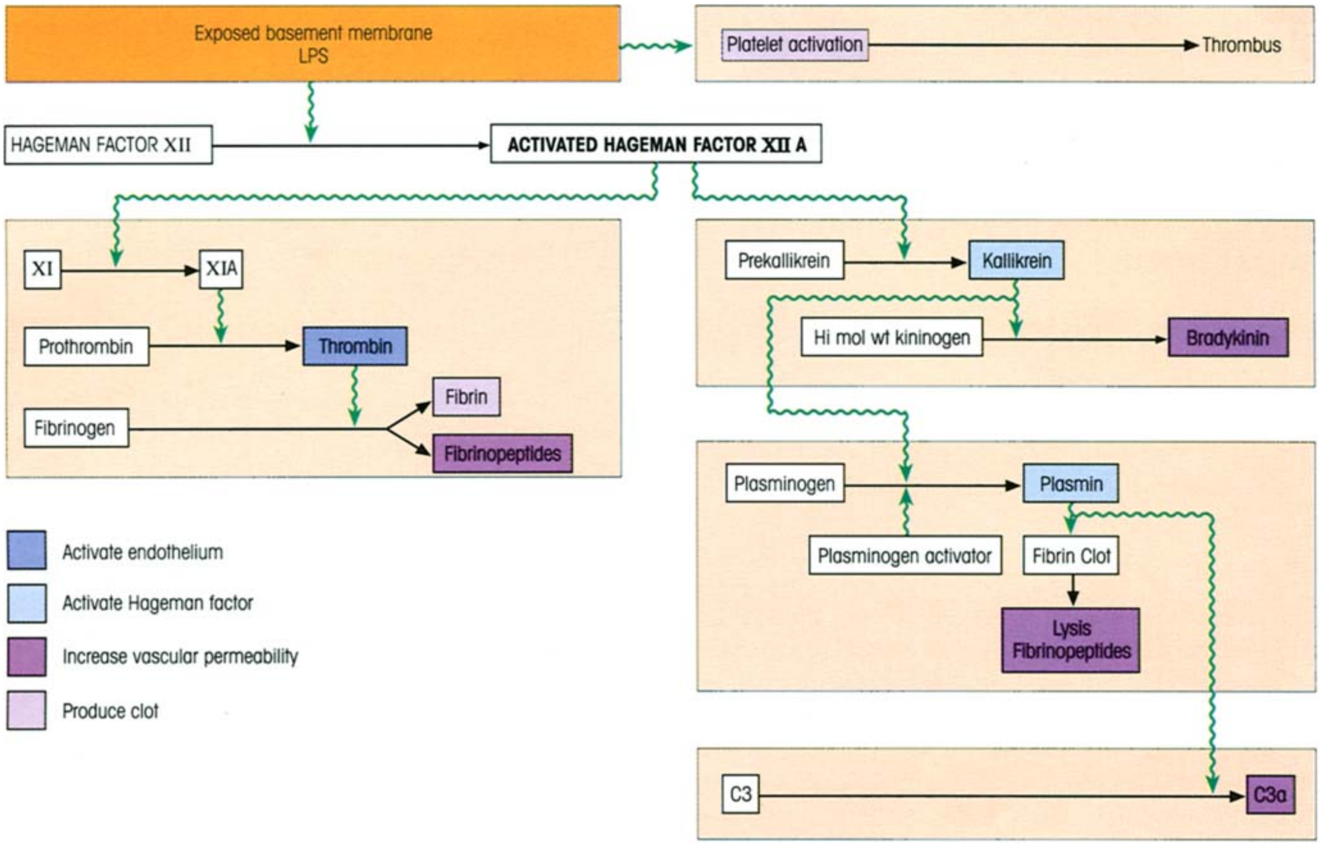

Figure 13.4. Events initiated by damaged vascular endothelium. Note that thrombin can also release platelet activating factor (PAF) which induces thrombus formation.

inhibitor, C4b-binding protein, the C3 control protein factors H and I, complement receptor CR1, *d*ecay *a*ccelerating *f*actor (DAF), *m*embrane *c*ofactor *p*rotein (MCP) and immunoconglutinin, and finally the proteins which block the membrane attack complex, homologous restriction factor and CD59 (discussed further on p. 307). Some of the acute phase proteins derived from the plasma transudate would be expected to act as protease inhibitors.

At the cellular level, PGE_2, transforming growth factor-β (TGFβ) and glucocorticoids are powerful regulators. PGE_2 is a potent inhibitor of lymphocyte proliferation and cytokine production by T-cells and macrophages. TGFβ deactivates macrophages by inhibiting the production of reactive oxygen intermediates and downregulating MHC class II expression; it also quells the cytotoxic enthusiasm of both macrophages and γ-interferon (IFNγ)-activated NK cells. Endogenous glucocorticoids produced via the hypothalamic–pituitary–adrenal axis exert their anti-inflammatory effects both through the *repression* of a number of genes, including those for proinflammatory cytokines and adhesion molecules, and the *induction* of the inflammation inhibitors lipocortin-1, secretory leukocyte proteinase inhibitor and IL-1 receptor antagonist. IL-10 inhibits antigen presentation, cytokine production and nitric oxide (NO) killing by macrophages, the latter inhibition being greatly enhanced by synergistic action with IL-4 and TGFβ.

Once the inflammatory agent has been cleared, these regulatory processes will normalize the site. When the inflammation traumatizes tissues through its intensity and extent, TGFβ plays a major role in the subsequent wound healing by stimulating fibroblast division and the laying down of new extracellular matrix elements.

Chronic inflammation

If an inflammatory agent persists, either because of its resistance to metabolic breakdown or through the inability of a deficient immune system to clear an infectious microbe, the character of the cellular response changes. The site becomes dominated by macrophages with varying morphology: many have an activated appearance, some form arrays of what are termed 'epithelioid' cells and others fuse to form giant cells. If an adaptive immune response is involved, lymphocytes in various guises will also be present. This characteristic **granuloma** walls off the persisting agent from the remainder of the body (see section on type IV hypersensitivity in Chapter 16, p. 343).

EXTRACELLULAR BACTERIA SUSCEPTIBLE TO KILLING BY PHAGOCYTOSIS AND COMPLEMENT

Bacterial survival strategies

The variety and ingenuity of these escape mechanisms are most intriguing and, as with virtually all infectious agents, if you can think of a possible avoidance strategy, some microbe will already have used it.

Evading phagocytosis

The cell walls of bacteria are multifarious (figure 13.5) and in some cases are inherently resistant to a number of microbicidal agents; but a common mechanism by which virulent forms escape phagocytosis is by synthesis of an outer **capsule**, which does not adhere readily to phagocytic cells and covers carbohydrate molecules on the bacterial surface which could otherwise be recognized by phagocyte receptors (figure 13.6a). For example, as few as 10 encapsulated pneumococci can kill a mouse but, if the capsule is removed by treatment with hyaluronidase, 10 000 bacteria are required for the job. Many pathogens evolve capsules which physically prevent access of phagocytes to C3b deposited on the bacterial cell wall.

Other organisms have actively **antiphagocytic** cell surface molecules and some go so far as to secrete **exotoxins**, which actually poison the leukocytes (figure 13.6b). Yet another devious ruse is to exploit binding to the surface of a nonphagocytic cell so gaining entry into a shelter from the depredations of the professional phagocyte (figure 13.6c). Presumably, some organisms try to avoid undue provocation of phagocytic cells by adhering to and *colonizing the external mucosal surfaces* of the intestine.

Challenging the complement system

Poor activation of complement. Normal mammalian cells are protected from complement destruction by regulatory proteins such as complement receptor CR1, MCP and DAF, which cause C3 convertase breakdown (see p. 307 for further discussion). Microorganisms lack these regulatory proteins so that, even in the absence of antibody, most of them would activate the alternative C pathway by stabilization of the $\overline{C3bBb}$ convertase on their surfaces. However, bacterial capsules in general tend to be poor activators of the alternative complement pathway and selective pressures have obviously favored the synthesis of capsules whose surface components do not favor stable binding of the convertase complex (figure 13.6d).

Acceleration of complement breakdown. Members of the *r*egulators of *c*omplement *a*ctivation (RCA) family which diminish C3 convertase activity include C4b-binding protein (C4BP), factor H and factor H-like protein 1 (FHL-1). Certain bacterial surface molecules, notably those rich in sialic acid, bind factor H (figure 13.6e), which then acts as a focus for the degradation of

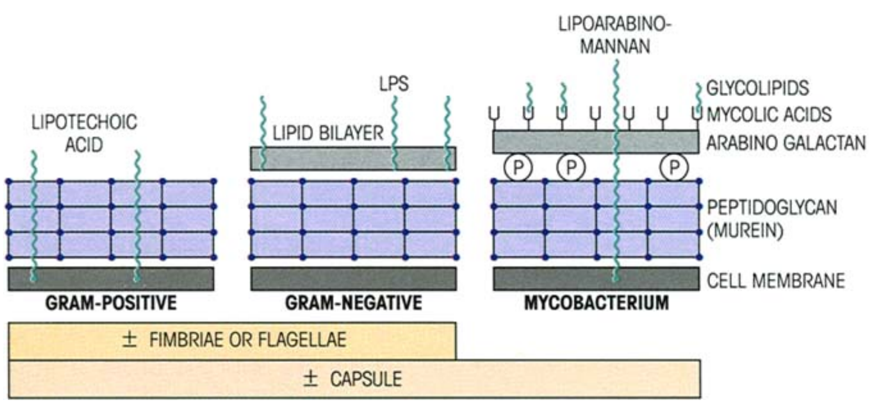

Figure 13.5. The structure of bacterial cell walls. All types have an inner cell membrane and a peptidoglycan wall which can be cleaved by lysozyme and lysosomal enzymes. The outer lipid bilayer of Gram-negative bacteria, which is susceptible to the action of complement or cationic proteins, sometimes contains lipopolysaccharide (LPS; also known as endotoxin; composed of O-specific oligosaccharide side-chains attached to a basal core polysaccharide, itself linked to the mitogenic moiety, lipid A; 148 O antigen variants of *Escherichia coli* are known). The mycobacterial cell wall is highly resistant to breakdown. When present, outer capsules may protect the bacteria from phagocytosis.

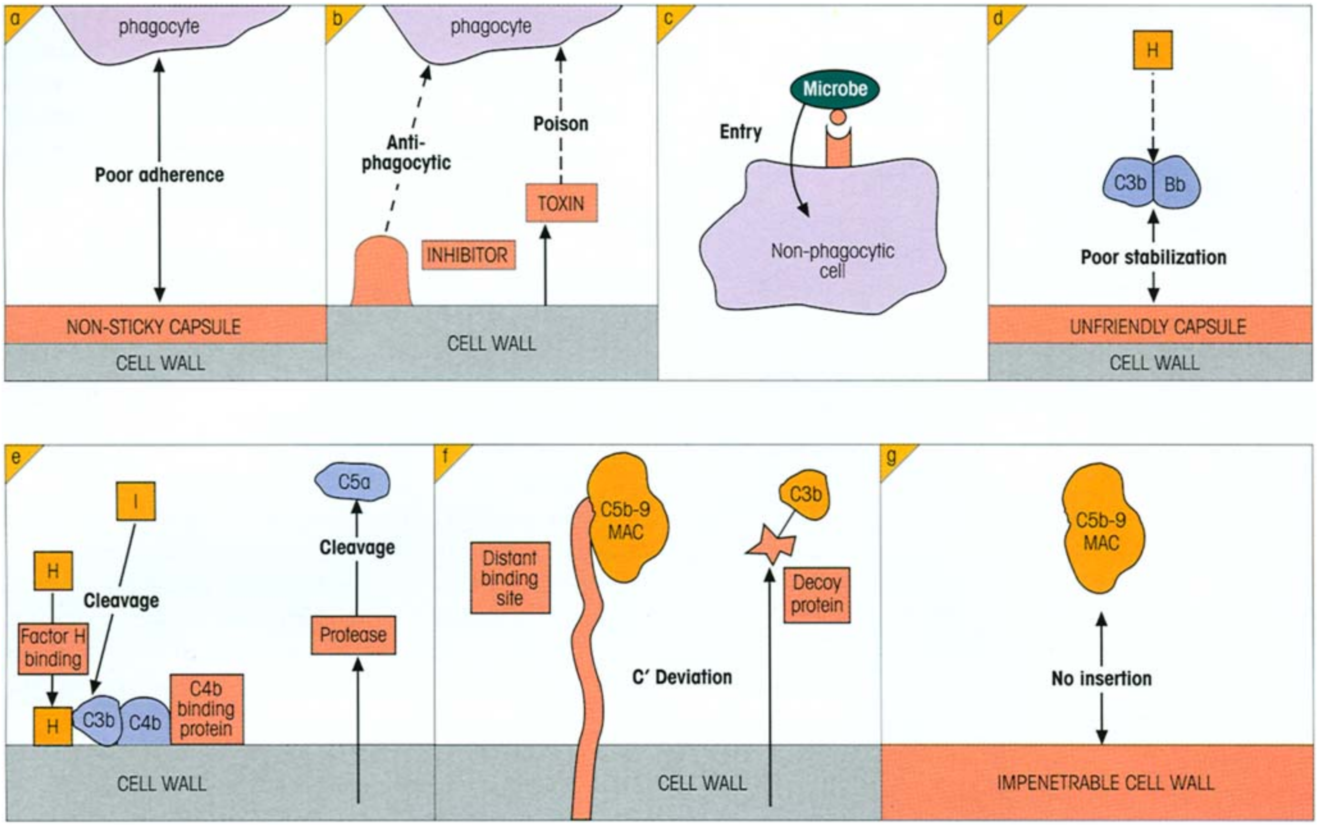

Figure 13.6. Avoidance strategies by extracellular bacteria. (a) Capsule gives poor phagocyte adherence. (b) Exotoxin poisons phagocyte. (c) Microbe attaches to surface component to enter nonphagocytic cell. (d) Capsule provides nonstabilizing surface for alternative pathway convertase. (e) Accelerating breakdown of complement by action of microbial products. (f) Complement effectors are deviated from the microbial cell wall. (g) Cell wall impervious to complement membrane attack complex (MAC).

C3b by the serine protease factor I (cf. p. 11). This is seen, for example, with *Neisseria gonorrhoeae*. Similarly, the hypervariable regions of the M-proteins of certain *Streptococcus pyogenes* (group A streptococcus) strains are able to bind FHL-1, whilst other strains downregulate complement activation by interacting with C4BP, this time acting as a cofactor for factor I-mediated degradation of the C4b component of the classical pathway C3 convertase $C\overline{4b2a}$. There is evidence that C4BP can also inhibit activation of the alternative pathway. Certain strains of group B streptococci produce a C5a-ase which may act as a virulence factor by proteolytically cleaving and thereby inactivating C5a (figure 13.6e).

Complement deviation. Some species manage to avoid lysis by deviating the complement activation site either to a secreted decoy protein or to a position on the bacterial surface distant from the cell membrane (figure 13.6f).

Resistance to insertion of terminal complement components. Gram-positive organisms (cf. figure 13.5) have evolved thick peptidoglycan layers which prevent the insertion of the lytic C5b–9 membrane attack complex into the bacterial cell membrane (figure 13.6g). Many capsules do the same.

Interfering with internal events in the macrophage

Enteric Gram-negative bacteria in the gut have developed a number of ways of influencing macrophage activity, including inducing apoptosis, enhancing the production of IL-1, preventing phagosome–lysosome fusion and affecting the actin cytoskeleton (figure 13.7).

Antigenic variation

Although the strategy of varying individual antigens in the face of a determined host antibody response is more usually associated with viruses and parasites, there are a few well-defined examples in bacteria. These include variation of surface lipoproteins in the lyme disease spirochete *Borrelia burgdorferi*, alterations in enzymes involved in synthesizing surface struc-

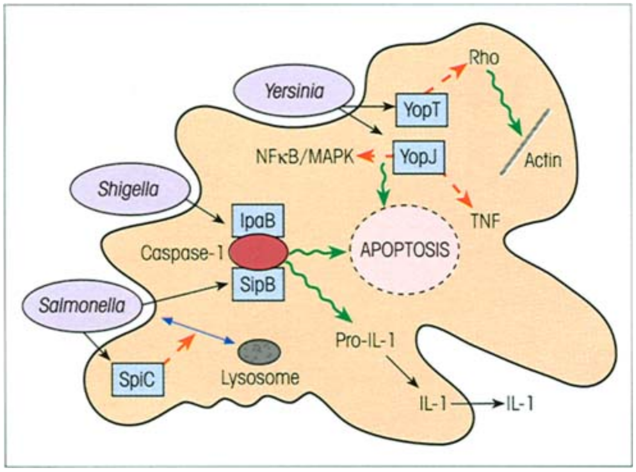

Figure 13.7. Evasion of macrophage defenses by enteric bacteria. The IpaB and SipB proteins secreted by *Shigella* and *Salmonella*, respectively, can activate caspase 1 and thereby set off a train of events that will lead to the death of the macrophage by apoptosis. Activated caspase 1 also triggers a protease which cleaves pro-IL-1, thereby causing the release of large amounts of this proinflammatory cytokine from the macrophage. Paradoxically, this may be advantageous to the bacteria because the subsequent migration of neutrophils to the intestinal lumen results in a loosening of intercellular junctions between the enterocytes, permitting cellular invasion of the basolateral surface by organisms from the lumen. The SpiC protein from *Salmonella* inhibits the trafficking of cellular vesicles, and therefore is able to prevent lysosomes fusing with phagocytic vesicles. *Yersinia* produces a number of Yop molecules (*Y*ersinia *o*uter *p*roteins) able to interfere with the normal functioning of the phagocyte. For example, YopJ inhibits TNF production and downregulates NFκB and MAP kinases, thereby facilitating apoptosis by inhibiting anti-apoptotic pathways. YopT prevents phagocytosis by modifying the GTPase Rho involved in regulating the actin cytoskeleton. (Based on Donnenberg M.S. (2000) *Nature* **406**, 768.)

tures in *Campylobacter jejuni* and antigenic variation of the pili in *Neisseria meningitidis*. In addition, new strains can arise, as has occurred with the life-threatening *E. coli* O157 : H7 which can cause hemolytic uremic syndrome and appears to have emerged about 50 years ago by incorporation of *Shigella* toxin genes into the *E. coli* 055 genome.

The host counter-attack

The defense mechanisms exploit the specificity and variability of the antibody molecule. Antibodies can defeat these devious attempts to avoid engulfment by neutralizing the antiphagocytic molecules and by binding to the surface of the organisms to focus the site for fixation of complement, so 'opsonizing' them for ingestion by neutrophils and macrophages or preparing them for the terminal membrane attack complex (Milestone 13.1). However, antibody production by B-cells usually requires T-cell help, and the T-cells need to be activated by antigen-presenting cells.

As already discussed in Chapter 1, *p*athogen-*a*ssociated *m*olecular *p*atterns (PAMPs), such as the all-important lipopolysaccharide (LPS) endotoxin of Gram-negative bacteria, peptidoglycan, lipoteichoic acids, mannans, bacterial DNA, double-stranded RNA and glucans, are all molecules which are broadly expressed by microbial pathogens but are not present on the host tissues. Thus these molecules serve as an

Milestone 13.1 — The Protective Effects of Antibody

The pioneering research which led to the recognition of the antibacterial protection afforded by antibody clustered in the last years of the 19th century. A good place to start the story is the discovery by Roux and Yersin in 1888, at the still famous Pasteur Institute in Paris, that the exotoxin of diphtheria bacillus could be isolated from a bacterium-free filtrate of the medium used to culture the organism. von Behring and Kitasato at Koch's Institute in Berlin in 1890 then went on to show that animals could develop an immunity to such toxins which was due to the development of specific neutralizing antidotes referred to generally as **antibodies**. They further succeeded in passively transferring immunity to another animal with serum containing the antitoxin. The dawning of an era of serotherapy came in 1894 with Roux's successful treatment of patients with diphtheria by injection of immune horse serum.

The immunological community then got its intellectual underwear in something of a twist over the next few years, first by advocates of the view that all bacterial immunity was due to antitoxins, and second by Metchnikoff's rigid espousal of phagocytosis itself as the main, if not the only, real bulwark of defense against infection. The situation was resolved by our shining knight Sir Almroth Wright in London in 1903 who proposed that the main action of the increased antibody produced after infection was to reinforce killing by the phagocytes. He called the antibodies **opsonins** (Gk. *opson*, a dressing or relish), because they prepared the bacteria as food for the phagocytic cells, and amply verified his predictions by showing that antibodies dramatically increased the phagocytosis of bacteria *in vitro*, thereby cleverly linking *innate* to *adaptive* immunity.

(*continued*)

Figure M13.1.1. Emil von Behring (1854–1917).

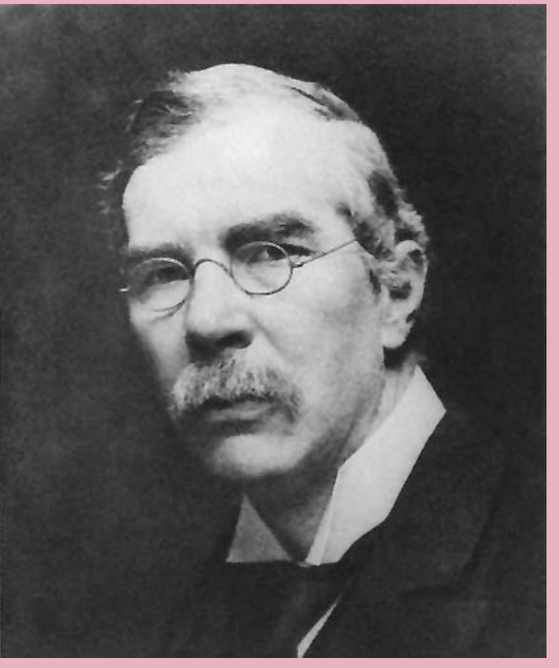

Figure M13.1.3. Sir Almroth Wright. (Slide kindly supplied by The Wellcome Centre Medical Photographic Library, London.)

Figure M13.1.2. von Behring extracting serum using a tap. Caricature by Lustigen Blättern, 1894. (Legend: 'Serum direct from the horse! Freshly drawn.') (Slide kindly supplied by The Wellcome Centre Medical Photographic Library, London.)

It was a feature of the time that major controversies were slugged out in pamphlets, books and plays, and it was impressive that Bernard Shaw should enter the fray on the side of Almroth Wright in his play *The Doctor's Dilemma*. In the preface he gave an evocative description of the function of opsonins: 'Sir Almroth Wright, following up one of Metchnikoff's most suggestive biological romances, discovered that the white corpuscles or phagocytes which attack and devour disease germs for us do their work only when we butter the disease germs appetizingly for them with a natural sauce which Sir Almroth named opsonins . . . .' (More extended and very readable accounts of immunology at the turn of the 19th century may be found in Humphrey J.H. & White R.G. (1970) *Immunology for Students of Medicine*, 3rd edn, Ch. 1. Blackwell Scientific Publications, Oxford; Craps L. (1993) *The Birth of Immunology*. Sandoz, Basel; and Silverstein A.M. (1989) *A History of Immunology*. Academic Press, San Diego.)

alerting service for the immune system, which detects their presence using a number of pattern recognition receptors (PRRs) expressed on the surface of antigen-presenting cells. Such receptors include the mannose receptor (CD206) which facilitates phagocytosis of microorganisms by macrophages and the scavenger receptor (CD204) which mediates clearance of bacteria from the circulation. Binding of LPS to the CD14 PRR on monocytes, macrophages, dendritic cells and B-cells leads to the recruitment of the toll-like receptor 4 (TLR4) molecule which is able to mediate signals activating the expression of a broad range of pro-inflammatory genes, including those for IL-1, IL-6, IL-12 and TNF and for the B7.1 (CD80) and B7.2

(CD86) costimulatory molecules. A related receptor, TLR2, recognizes Gram-positive bacterial cell wall components.

Toxin neutralization

Circulating antibodies act to neutralize the soluble antiphagocytic molecules and other exotoxins (e.g. phospholipase C of *Clostridium welchii*) released by bacteria. Combination near the biologically active site of the toxin would stereochemically block reaction with the substrate, particularly if it were macromolecular; combination distant from the active site may also cause inhibition through allosteric conformational changes. In its complex with antibody, the toxin may be unable to diffuse away rapidly and will be susceptible to phagocytosis, especially if the complex can be increased in size by the action of naturally occurring autoantibodies to complexed IgG (antiglobulin factors) and C3b (immunoconglutinin, not to be confused with bovine *conglutinin*, a nonantibody molecule which combines with the carbohydrate portion of C3b).

Opsonization of bacteria

Independently of antibody. Differences between the carbohydrate structures on bacteria and self are exploited by the **collectins**, a series of molecules with similar ultrastructure to C1q and which bear C-terminal lectin domains. These include mannose-binding protein (MBP; also referred to as mannose-binding lectin, MBL), lung surfactant proteins SP-A and SP-D and, in cattle, conglutinin, which all recognize carbohydrate ligands. Mannose-binding protein can bind to terminal mannose on the bacterial surface and then interacts with MBP-associated serine protease (MASP) which is homologous in structure to C1r and C1s. Thus the interaction is closely similar to that of C1q with C1r and s (cf. p. 22) and, in this way, leads to the antibody-independent activation of the classical pathway. SP-A, SP-D and conglutinin can all act as opsonins (see Milestone 13.1) and can mediate phagocytosis by virtue of their binding to the C1q receptor.

Augmented by antibody. Encapsulated bacteria which resist phagocytosis become extremely attractive to neutrophils and macrophages when coated with antibody and their rate of clearance from the bloodstream is strikingly enhanced (figure 13.8). The less effective removal of coated bacteria in complement-depleted animals emphasizes the synergism between antibody and complement for opsonization which is mediated

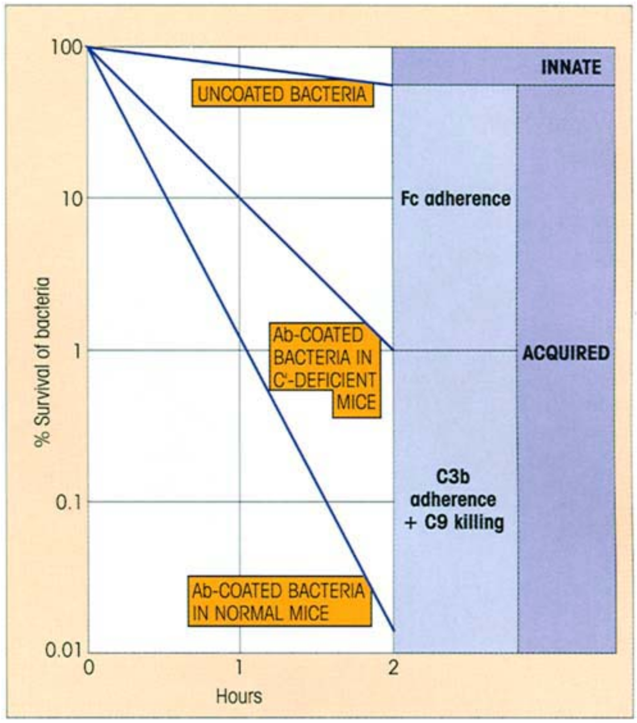

Figure 13.8. Effect of opsonizing antibody and complement on rate of clearance of virulent bacteria from the blood. The uncoated bacteria are phagocytosed rather slowly (*innate immunity*) but, on coating with antibody, adherence to phagocytes is increased many-fold (*acquired immunity*). The adherence is less effective in animals temporarily depleted of complement. This is a hypothetical but realistic situation; the natural proliferation of the bacteria has been ignored.

through specific high affinity receptors for IgG and C3b on the phagocyte surface (figure 13.9). It is clearly advantageous that the subclasses which bind strongly to these Fc receptors (e.g. IgG1 and IgG3 in the human) also fix complement well, it being appreciated that the heterodimer of C3b bound to IgG is a very efficient opsonin because it engages two receptors simultaneously. Complexes containing C3b and C4b may show immune adherence to the CR1 complement receptors on erythrocytes to provide aggregates which are transported to the liver and spleen for phagocytosis.

Some elaboration on **complement receptors** may be pertinent at this stage. The CR1 receptor (CD35) for C3b is also present on neutrophils, eosinophils, monocytes, B-cells and lymph node follicular dendritic cells. Together with the CR3 receptor (CD11b/CD18), it has the main responsibility for clearance of complexes containing C3. The *CR1* gene is linked in a cluster with C4b-binding protein and factor H, all of which subserve a regulatory function by binding to C3b or C4b to disassemble the C3/C5 convertases, and act as cofactors for the proteolytic inactivation of C3b and C4b by factor I.

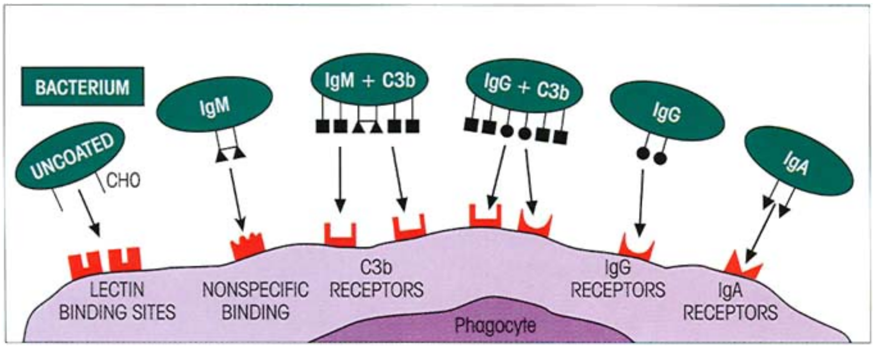

Figure 13.9. Immunoglobulin and complement coats greatly increase the adherence of bacteria (and other antigens) to macrophages and neutrophils. Uncoated bacteria adhere to lectin-like sites, including the mannose-binding receptor. There are no specific binding sites for IgM (▲▲), but there are high affinity receptors for IgG (Fc) (●) and iC3b (■) on the macrophage surface which considerably enhance the strength of binding. The augmenting effect of complement is due to the fact that two adjacent IgG molecules can fix many C3b molecules, thereby increasing the number of links to the macrophage (cf. 'bonus effect of multivalency'; p. 87). Although IgM does not bind specifically to the macrophage, it promotes adherence through complement fixation. Specific receptors for the Fcα domains of IgA have also been defined.

CR2 receptors (CD21) for iC3b, C3dg and C3d are present on B-cells and follicular dendritic cells and transduce accessory signals for B-cell activation especially in the germinal centers (cf. p. 190). Their affinity for the Epstein–Barr virus (EBV) provides the means for entry of the virus into the B-cell.

CR3 receptors (CD11b/CD18 on neutrophils, eosinophils, monocytes and NK cells) bind iC3b, C3dg and C3d. They are related to LFA-1 and CR4 (CD11c/CD18, binds iC3b and C3dg) in being members of the β_2 integrin subfamily (cf. table 8.2). CR5 is found on neutrophils and platelets and binds C3d and C3dg. A number of other complement receptors have been described including some with specificity for C1q, for C3a and C4a, and the CD88 molecule with specificity for C5a.

Some further effects of complement

Some strains of Gram-negative bacteria which have a lipoprotein outer wall resembling mammalian surface membranes in structure are susceptible to the bactericidal action of fresh serum containing antibody. The antibody initiates the development of a complement-mediated lesion which is said to allow access of serum lysozyme to the inner peptidoglycan wall of the bacterium to cause eventual cell death. Activation of complement through union of antibody and bacterium will also generate the C3a and C5a anaphylatoxins leading to extensive transudation of serum components, including more antibody, and to the chemotactic attraction of neutrophils to aid in phagocytosis, as described earlier under the acute inflammation umbrella (cf. figures 2.18 and 13.3).

The secretory immune system protects the external mucosal surfaces

We have earlier emphasized the critical nature of the mucosal barriers, particularly in the gut, where there is a potentially hostile interface with the microbial hordes. With an area of around 400 m^2, give or take a tennis court or two, the epithelium of the adult mucosae represents the most frequent portal of entry for common infectious agents, allergens and carcinogens. The need for well-marshaled, highly effective mucosal immunity is glaringly obvious. Awareness of such a need was evident even in bygone days. Per Brandtzaeg relates an amusing historical example.

In ancient times, kings were perpetually worried that they would be poisoned by their enemies. (One heard tell that the habit of clinking glasses together in a toast has its origin in the custom of pouring some of one's wine into the glass of the next person and so on until the operation had come full circle.) Anyway, Mithridates VI Eupator, King of Pontus (now part of Turkey) in the 1st century BC, was so frightened of being poisoned that he attempted to increase his intestinal defenses by drinking the blood of ducks which had been fed poisonous weeds. After defeat by the Romans and mutiny in his army he tried unsuccessfully to poison himself, testimony to the effectiveness of oral immunization. As a grisly negative control, his two daughters had no difficulty in committing suicide using the same poison, whereupon the desperate king was forced to order his bodyguard to kill him with a more conventional weapon.

The gut mucosal surfaces are defended by both antigen-specific and nonantigen-specific mechanisms.

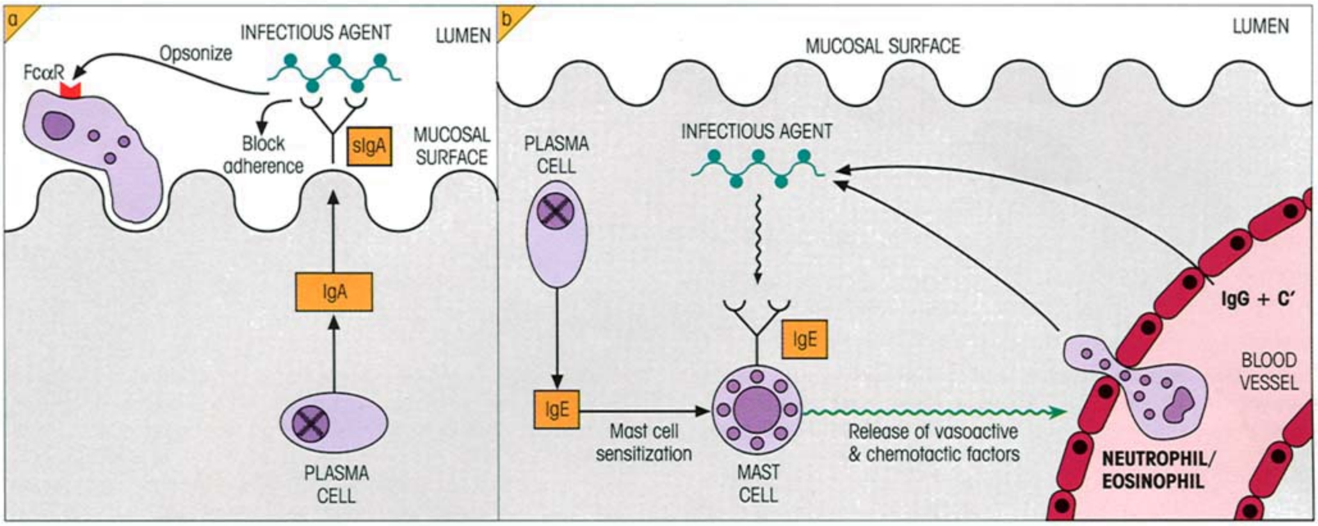

Figure 13.10. Defense of the mucosal surfaces. (a) IgA opsonizes organisms and prevents adherence to the mucosa. (b) IgE recruits agents of the immune response by firing the release of mediators from mast cells.

Among the nonspecific mechanisms, those provided by antimicrobial peptides are worth highlighting. These are produced not only by neutrophils and macrophages but also by mucosal epithelium, where they serve as a primary bacterial defense barrier. As described in Chapter 1, the group of antimicrobial peptides called defensins cause lysis of bacteria via disruption of their surface membranes. Specific immunity is provided by secretory IgA and IgM, with IgA1 predominating in the upper areas and IgA2 in the large bowel. Most other mucosal surfaces are also protected predominantly by IgA with the surprising exception of the reproductive tract tissues of both male and female, where the dominant antibody isotype is IgG. The size of the task is highlighted by the fact that 80% of the Ig-producing B-cells in the body are present in the secretory mucosae and exocrine glands. IgA antibodies afford protection in the external body fluids, tears, saliva, nasal secretions and those bathing the surfaces of the intestine and lung by coating bacteria and viruses and preventing their adherence to the epithelial cells of the mucous membranes, which is essential for viral infection and bacterial colonization. Secretory IgA molecules themselves have very little innate adhesiveness for epithelial cells, but high affinity Fc receptors for this Ig class are present on macrophages and neutrophils and can mediate phagocytosis (figure 13.10a).

If an infectious agent succeeds in penetrating the IgA barrier, it comes up against the next line of defense of the secretory system (see p. 154) which is manned by IgE. There are obvious parallels between the ways in which complement-derived anaphylatoxins and IgE utilize the mast cell to cause local amplification of the immune defenses. It is worth noting that most serum IgE arises from plasma cells in mucosal tissues and in the lymph nodes that drain them. Although present in low concentration, IgE is bound very firmly to the Fc receptors of the mast cell (see p. 55) and contact with antigen leads to the release of mediators which effectively recruit agents of the immune response and generate a local acute inflammatory reaction. Thus histamine, by increasing vascular permeability, causes the transudation of IgG and complement into the area, while chemotactic factors for neutrophils and eosinophils attract the effector cells needed to dispose of the infectious organism coated with specific IgG and C3b (figure 13.10b). Engagement of the Fcγ and C3b receptors on local macrophages by such complexes will lead to secretion of factors which further reinforce these vascular permeability and chemotactic events. Broadly, one would say that immune exclusion in the gut is noninflammatory, but immune elimination of organisms which penetrate the mucosa is proinflammatory.

Where the opsonized organism is too large to be engulfed, phagocytes can kill by an extracellular mechanism after attachment by their Fcγ receptors. This phenomenon, termed *a*ntibody-*d*ependent *c*ellular *c*ytotoxicity (ADCC), has been discussed earlier (see p. 32) and there is evidence for its involvement in parasitic infections (see p. 271).

The mucosal tissues contain a variety of T-lymphocyte species, but their role and that of the mucosal epithelial cells, other than in a helper function for local antibody production, is of less relevance for the defense against extracellular bacteria.

Some specific bacterial infections

First let us see how these considerations apply to defense against infection by common organisms such as streptococci and staphylococci. β-Hemolytic **streptococci** were classified by Lancefield according to their carbohydrate antigen, and the most important for human disease are those belonging to group A. *Streptococcus pyogenes* posed a major worldwide health threat; in 1981, more than six million children suffered from associated rheumatic disease in India alone, and resurgence of a highly virulent strain causing rheumatic fever and toxic shock syndrome has occurred in the USA.

The most important virulence factor is the surface M-protein (variants of which form the basis of the Griffith typing). This protein is an acceptor for factor H which facilitates C3b breakdown, and binds fibrinogen and its fragments which cover sites that may act as complement activators (figure 13.11). It thereby inhibits opsonization and the protection afforded by antibodies to the M-component is attributable to the striking increase in phagocytosis which they induce. The ability of group A streptococci to elicit cross-reactive autoantibodies which bind to cardiac myosin is thought to be involved in poststreptococcal autoimmune disease. High titer antibodies to the streptolysin O exotoxin (ASO), which damages membranes, are indicators of recent streptococcal infection. The *s*treptococcal *p*yrogenic *e*xotoxins SPE A, B and C are superantigens associated with scarlet fever and toxic shock syndrome. The toxins are neutralized by antibody and the erythematous intradermal reaction to injected toxin (the Dick reaction) is only seen in individuals lacking antibody. Antibody can also neutralize bacterial enzymes like hyaluronidase which act to spread the infection.

Streptococcus mutans is an important cause of both endocarditis and dental caries. The organism has a constitutive enzyme, glucosyltransferase, able to convert sucrose to dextran which is utilized for adhesion to the tooth surface. Passive transfer of IgG, but not IgA or IgM, antibodies to *S. mutans* in monkeys conferred protection against caries. It is thought that IgG antibody and complement in the gingival crevicular fluid bathing the tooth opsonize the bacteria to facilitate phagocytosis and killing by neutrophils. Curiously, a single treatment with a monoclonal anti-*S. mutans* kept teeth free of the microorganism for 1 year, perhaps due to a shift in bacterial ecology, with the place vacated by *S. mutans* being filled with another organism from the oral flora.

Virulent forms of **staphylococci**, of which *Staphylococcus aureus* is perhaps the most common, resist phagocytosis. This may be due partly to capsule formation *in vivo* and partly to the elaboration of factors such as a coagulase enzyme which could protect the bacterium by a barrier of fibrin. It has been suggested that the ability of a cell wall component, protein A, to combine with the Fc portion of IgG (other than subclass IgG3) is responsible for inhibition of phagocytosis by virulent strains, but IgG–protein A complexes fix complement and one study reports that protein A actually increases complement-mediated phagocytosis. We must return an open verdict on that issue. It seems to be accepted that *S. aureus* is readily phagocytosed in the presence of *adequate* amounts of antibody, but a small proportion of the ingested bacteria survives and they are difficult organisms to eliminate completely. Where the infection is inadequately controlled, severe lesions may occur in the immunized host as a conse-

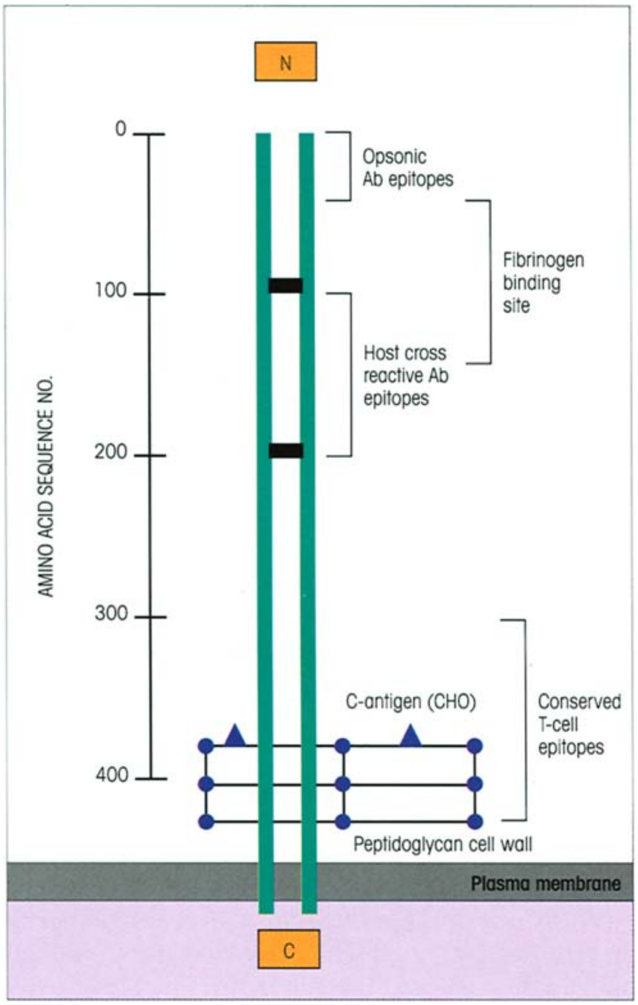

Figure 13.11. Streptococcal M-protein from *S. pyogenes*. The coiled-coil dimers are approximately 50 nm long. (Based on Robinson J.H. & Kehoe M.A. (1992) *Immunology Today* **13**, 362.)

quence of type IV delayed hypersensitivity reactions. Thus, staphylococci were found to be avirulent when injected into mice passively immunized with antibody, but caused extensive tissue damage in animals previously given sensitized T-cells.

Other examples where antibodies are required to overcome the inherently antiphagocytic properties of **bacterial capsules** are seen in immunity to infection by pneumococci, meningococci and *Haemophilus influenzae*. *Bacillus anthrax* possesses an antiphagocytic capsule composed of a γ-polypeptide of D-glutamic acid but, although anticapsular antibodies effectively promote uptake by neutrophils, the exotoxin is so potent that vaccines are inadequate unless they also stimulate antitoxin immunity. In addition to releasing such lethal exotoxins, *Pseudomonas aeruginosa* also produces an elastase that inactivates C3a and C5a; as a result, only minimal inflammatory responses are made in the absence of neutralizing antibodies.

The ploy of **diverting complement activation** to insensitive sites is seen rather well with different strains of Gram-negative *Salmonella* and *Escherichia coli* organisms which vary in the number of O-specific oligosaccharide side-chains attached to the lipid-A-linked core polysaccharide of the endotoxin (cf. figure 13.5). Variants with long side-chains are relatively insensitive to killing by serum through the alternative complement pathway (see p. 11); as the side-chains become shorter and shorter, the serum sensitivity increases. Although all variants activate the alternative pathway, only those with short or no side-chains allow the cytotoxic membrane attack complex to be inserted near to the outer lipid bilayer (figure 13.6f). On the other hand, antibodies focus the complex to a more vulnerable site.

The destruction of gonococci by serum containing antibody is dependent upon the formation of the membrane attack complex, and rare individuals lacking C8 or C9 are susceptible to *Neisseria* infection. *N. gonorrhoeae* (gonococci) specifically binds complement proteins and prevents their insertion in the outer membranes, but antibody, like a ubiquitous 'Mr Fixit', corrects this situation, at least so far as the host is concerned. With respect to the infective process itself, IgA produced in the genital tract in response to these organisms inhibits the attachment of the bacteria, through their pili, to mucosal cells, but seems unable to afford adequate protection against reinfection. This seems to be due to a very effective antigenic shift mechanism which alters the sequence of the expressed pilin by gene conversion. The outer membrane protein P or B inhibits phagocyte microbicidal activity and induces apoptosis in the target cells. Failure to achieve good protection might also be a reflection of the ability of the gonococci to produce a protease which cleaves a proline-rich sequence present in the hinge region of IgA1 (but not of IgA2), although the presence in most individuals of neutralizing antibodies against this protease may interfere with its proteolytic activity. Meningococci, which frequently infect the nasopharynx, *H. influenzae* and *S. pneumoniae* have similar unfair proteases.

Cholera is caused by the colonization of the small intestine by *Vibrio cholerae* and the subsequent action of its enterotoxin. The B subunits of the toxin bind to specific GM1 monosialoganglioside receptors and translocate the A subunit across the membrane where it activates adenyl cyclase. The increased cAMP then causes fluid loss by inhibiting uptake of sodium chloride and stimulating active Cl^- secretion by intestinal epithelial cells. Locally synthesized IgA antibodies against *V. cholerae* lipopolysaccharide and the toxin provide independent protection against cholera, the first by inhibiting bacterial adherence to the intestinal wall, the second by blocking attachment of the toxin to its receptor. In accord with this analysis are the epidemiological data showing that children who drink milk with high titers of IgA antibodies specific for either of these antigens are less likely to develop clinical cholera.

Yersinia and *Salmonella* are among the select number of bacterial pathogens which have evolved special mechanisms to enter, survive and replicate within normally **nonphagocytic host cells**. The former gains entry through binding of its outer membrane protein, invasin, to multiple β_1 integrin receptors on the host cell. *Salmonella*, on the other hand, induces membrane ruffling on the target cell, linked in some way to signaling events which lead to bacterial uptake.

The ways in which antibody can parry the different facets of bacterial invasion are summarized in figure 13.12.

BACTERIA WHICH GROW IN AN INTRACELLULAR HABITAT

Bacterial gambits

Some strains of bacteria, such as the tubercle and leprosy bacilli and *Listeria* and *Brucella* organisms, escape the wrath of the immune system by cheekily fashioning an intracellular life within one of its strongholds, the macrophage no less. Mononuclear phagocytes are a good target for such organisms in the sense that they are very mobile and allow wide dissemination throughout the body. Entry of opsonized bacteria is facilitated by phagocytic uptake after attachment to

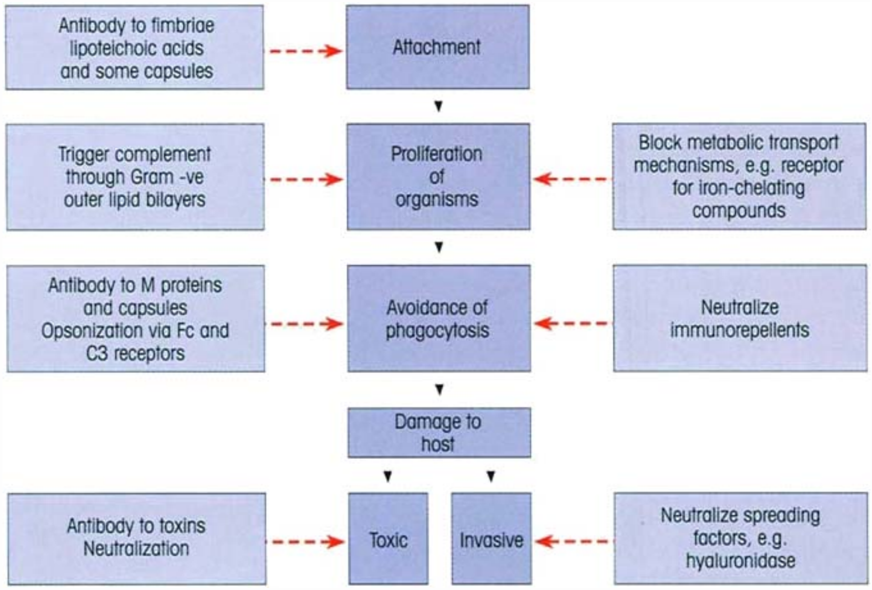

Figure 13.12. Antibody defenses against bacterial invasion.

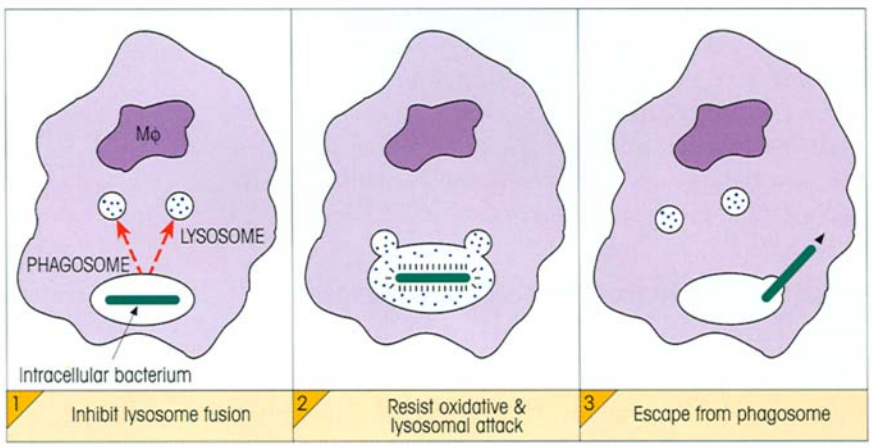

Figure 13.13. Evasion of phagocytic death by intracellular bacteria.

Fcγ and C3b receptors but, once inside, many of them defy the mighty macrophage by subverting the innate killing mechanisms in a variety of ways. Organisms such as *Mycobacterium tuberculosis* inhibit fusion of the lysosomes with the phagocytic vacuole containing the ingested bacterium (figure 13.13). Mycobacterial lipids, such as lipoarabinomannan, obstruct priming and activation of the macrophage and also protect the bacteria from attack by scavenging reactive oxygen intermediates such as superoxide anion, hydroxyl radicals and hydrogen peroxide (cf. p. 6). Organisms such as *Listeria monocytogenes* use a special lysin to escape from their phagosomal prison to lie happily free within the cytoplasm; some rickettsiae and the protozoan *Trypanosoma cruzi* can do the same. If there is a mechanism available to inhibit, some microorganisms will eventually find a way to do it.

Defense is by T-cell-mediated immunity (CMI)

In an elegant series of experiments, Mackaness demonstrated the importance of CMI reactions for the killing of intracellular parasites and the establishment of an immune state. Animals infected with moderate doses of *M. tuberculosis* overcome the infection and are immune to subsequent challenge with the bacillus. The immunity can be transferred to a normal recipient with T-lymphocytes but not macrophages or serum from an immune animal. Supporting this view, that specific immunity is mediated by T-cells, is the greater susceptibility to infection with tubercle and leprosy bacilli of mice in which the T-lymphocytes have been depressed by thymectomy plus anti-T-cell monoclonals, or in which the TCR genes have been disrupted by homologous gene recombination (knockout mice).

Activated macrophages kill intracellular parasites

When monocytes first settle down in the tissue to become 'resident' macrophages they are essentially downregulated with respect to the expression of surface receptors and function. They can be activated in several stages (figure 13.14), but the ability to kill obligate intracellular microbes only comes after stimulation by macrophage activating factor(s), such as IFNγ released from stimulated cytokine-producing T-cells. Foremost amongst the killing mechanisms which are upregulated are those mediated by reactive oxygen intermediates and NO·. The activated macrophage is undeniably a remarkable and formidable cell, capable of secreting the 60 or so substances which are concerned in chronic inflammatory reactions (figure 13.15)—not the sort to meet in an alley on a dark night!

The mechanism of T-cell-mediated immunity in the Mackaness experiments now becomes clear. Specifically primed T-cells react with processed antigen derived from the intracellular bacteria present on the surface of the infected macrophage in association with MHC II; the subsequent release of cytokines activates the macrophage and endows it with the ability to kill the organisms it has phagocytosed (figure 13.16).

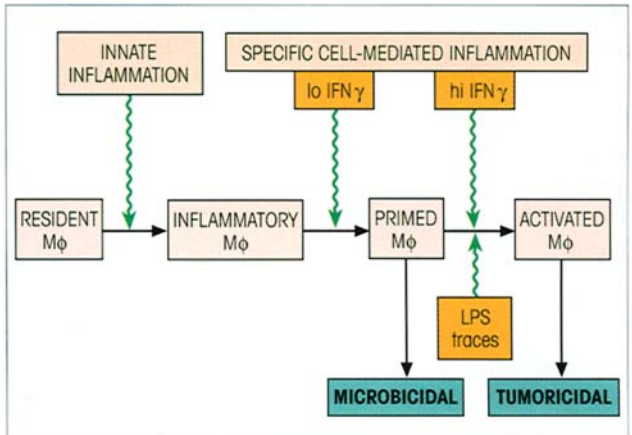

Figure 13.14. Stages in the activation of macrophages (Mϕ) for microbicidal and tumoricidal function. Macrophages taken from sites of inflammation induced by complement or nonimmunological stimuli, such as thioglycollate, are considerably increased in size, acid hydrolase content, secretion of neutral proteinases and phagocytic function. If we may give one example, the C3b receptors on resident Mϕ are not freely mobile in the membrane and so cannot permit the 'zippering' process required for phagocytosis (see p. 6); consequently, they bind but do not ingest C3b-coated red cells. Inflammatory Mϕ, on the other hand, have C3 receptors which display considerable lateral mobility and the C3-opsonized erythrocytes are readily phagocytosed. In addition to the dramatic upregulation of intracellular killing mechanisms, striking changes in surface components accompany activation. In mouse macrophages there is an increase in class II MHC (dramatic), Fc receptors for IgG2b and binding sites for tumor cells; in contrast, the mannose receptor (CD206), the F4/80 marker and IgG2a receptors all decline, while the Mac-I (CD11b) component of the CR3 iC3b receptor remains unchanged.

Examples of intracellular bacterial infections

Listeria

Undoubtedly, T-helper-1 (Th1)-activated macrophages are crucially important for the ultimate elimination

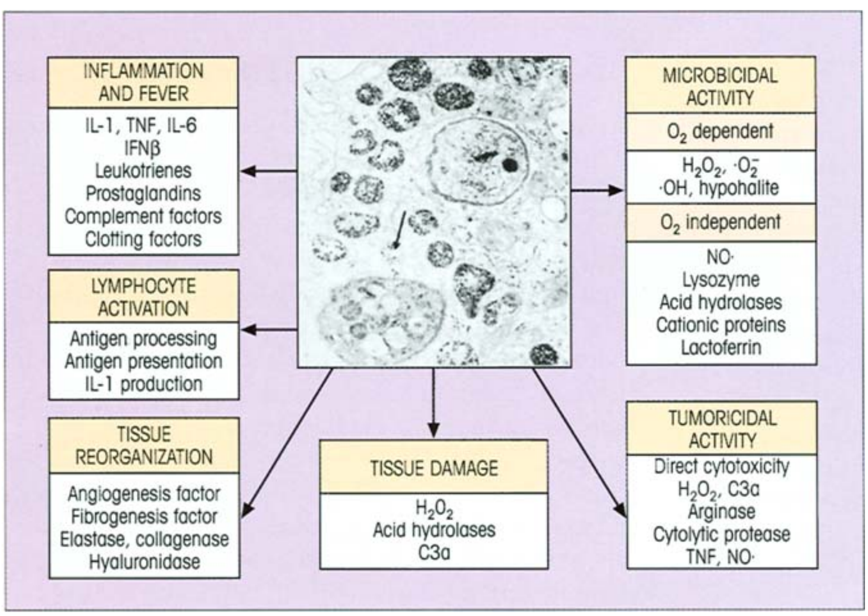

Figure 13.15. The role of the activated macrophage in the initiation and mediation of chronic inflammation with concomitant tissue repair, and in the killing of microbes and tumor cells. It is possible that macrophages differentiate along distinct pathways to subserve these different functions. The electron micrograph shows a highly activated macrophage with many lysozomal structures which have been highlighted by the uptake of thorotrast; one (arrowed) is seen fusing with a phagosome containing the protozoan *Toxoplasma gondii*. (Courtesy of Professor C. Jones.)

of intracellular *Listeria*, but the complex role of innate mechanisms early in infection must be considered. These are outlined in figure 13.17, where attention should be drawn to the bactericidal action of neutrophils and the central action of IL-12, which generates the macrophage activator IFNγ through its stimulation of NK cells and recruitment of Th1 helpers. Mutant mice lacking αβ and/or γδ T-cells reveal that these two cell types make comparable contributions to resistance against primary *Listeria* infection, but that the αβ TCR set bears the major responsibility for conferring protective immunity. γδ T-cells control the local tissue response at the site of microbial replication and γδ knockout mutants develop huge abscesses when infected with *Listeria*.

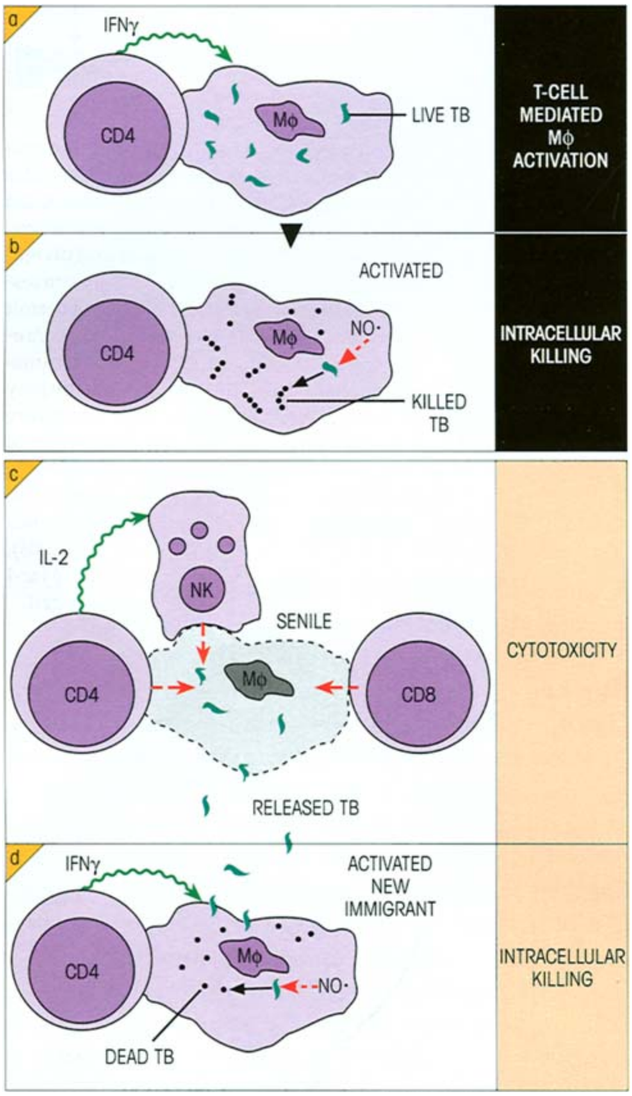

Figure 13.16. The 'cytokine connection': nonspecific murine macrophage killing of intracellular bacteria triggered by a specific T-cell-mediated immunity reaction. (a) Specific CD4 Th1 cell recognizes mycobacterial peptide associated with MHC class II and releases Mϕ activating IFNγ. (b) The activated Mϕ kills the intracellular TB, mainly through generation of toxic NO·. (c) A 'senile' Mϕ, unable to destroy the intracellular bacteria, is killed by CD8 and CD4 cytotoxic cells and possibly by IL-2-activated NK cells. The Mϕ then releases live tubercle bacilli which are taken up and killed by newly recruited Mϕ susceptible to IFNγ activation (d). Human monocytes require activation by both IFNγ and IL-4 plus a CD23-mediated signal for induction of iNO synthase and production of NO·.

Tuberculosis

Tuberculosis (TB) is on the rampage, aided by the emergence of multidrug-resistant strains of *Mycobacterium tuberculosis*. It is predicted that a total of 225 million new cases and 79 million deaths will occur between 1998 and 2030. This is not helped by the fact that human immunodeficiency virus (HIV)-infected individuals with low CD4 counts have increased susceptibility to *M. tuberculosis* infection.

With respect to host defense mechanisms, as seen with *Listeria* infection, murine macrophages activated by IFNγ can destroy intracellular mycobacteria with magisterial ease, largely through the generation of toxic NO·. Some parasitized macrophages reach a stage at which they are too incapacitated to be stirred into action by T-cell messages, and here a somewhat ruthless strategy has evolved in which the host deploys cytotoxic CD8, and possibly CD4 and NK cells, to execute the helpless macrophage and release the live mycobacteria; these should now be taken up by newly immigrant phagocytic cells susceptible to activation by IFNγ and summarily disposed of (figure 13.16).

The position is more complicated in the human. IFNγ-stimulated human monocytes cannot eliminate intracellular TB, although there is some stasis and IFNγ does upregulate expression of the 1-hydroxylase which converts vitamin D_3 into the 1,25-dihydroxy form, a potent activator of antimycobacterial mechanisms. However, there is no induction of NO· synthase (iNOS type II) unless IL-4 is also added to the system, whereupon there is copious production of NO· in a CD23-dependent reaction.

A further plethora of potential defensive mechanisms is emerging. The TB products Ag85B (a mycolyl transferase) and ESAT-6 are potent inducers of IFNγ from CD4 cells. $CD4^-8^-$ αβ T-cells proliferate in response to mycolic acid and lyse cells presenting the acid in the context of CD1b, while human $V\gamma_2V\delta_2$ T-cells recognize protein antigens, isopentenyl pyrophosphates and prenyl pyrophosphates from *M. tuberculosis*. The γδ T-cells have an eerie, almost compelling, relationship to mycobacterial antigens. A substantial proportion of them can lyse target cells presenting mycobacterial antigens and, in the mouse, a

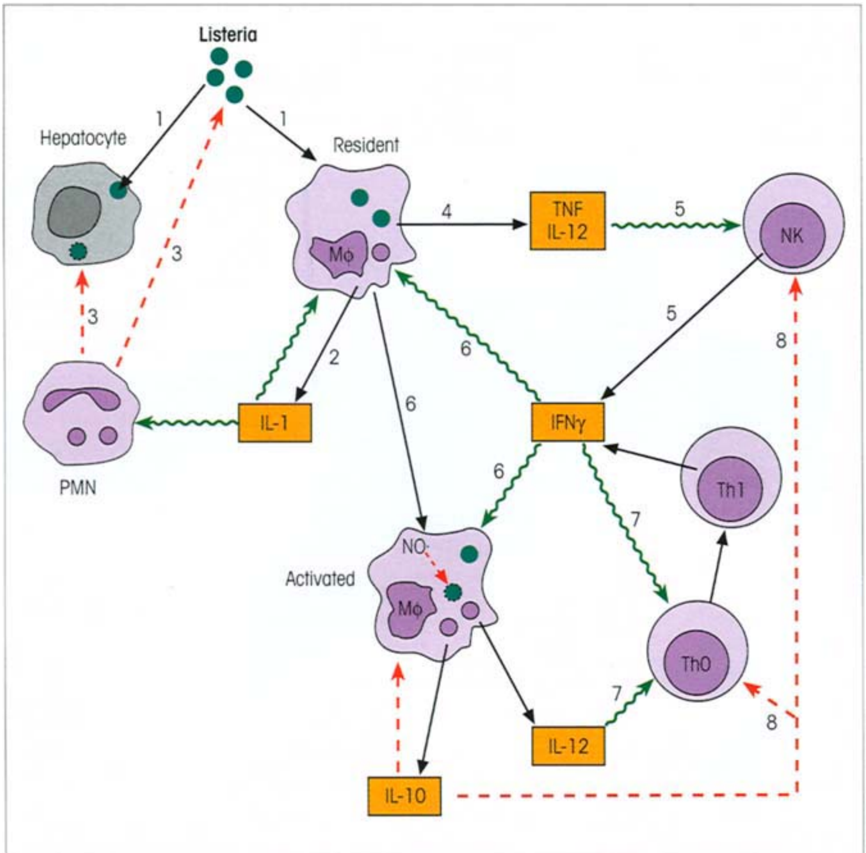

Figure 13.17. Immune response to *Listeria* infection. (1) *Listeria* infects resident macrophages and hepatocytes; (2) the Mϕ release IL-1 which activates polymorphs; (3) the activated PMN destroy *Listeria* bacilli by direct contact and are cytotoxic for infected hepatocytes; (4) the infected Mϕ release TNF and IL-12 which stimulate NK cells; (5) NK cells secrete IFNγ, which (6) activates the macrophage to produce NO· and kill intracellular *Listeria*; (7) IFNγ plus Mϕ-derived IL-12 recruit Th1 cells which reinforce Mϕ activation through the production of IFNγ. (8) Eventual synthesis of IL-10, encouraged by the action of immune complexes, downregulates Mϕ, NK and Th1 activity. (Based on an article by Rogers H.W., Tripps C.S. & Unanue E.R. (1995) *The Immunologist* **3**, 152.)

high percentage react to the 65 kDa heat-shock protein. A vital role for these cells in murine TB is indicated by the death of γδ TCR deletion mutants after an infection with *M. tuberculosis* which was still tolerated by immunocompetent animals.

Given these potential CMI defenses, why do some individuals fail to eradicate their infections with mycobacteria and other intracellular facultative bacteria and so suffer from diseases such as TB and leprosy even though they have established Th1 responses? One important clue is provided by the demonstration that inbred strains of mice differ dramatically in their susceptibility to infection by various mycobacteria, *Salmonella typhimurium* and *Leishmania donovani*. Resistance is associated with a T-cell-independent enhanced state of macrophage priming for bactericidal activity involving oxygen and nitrogen radicals. Moreover, macrophages from resistant strains have increased MHC class II expression and a higher respiratory burst, are more readily activated by IFNγ, and induce better stimulation of T-cells, whereas macrophages from susceptible strains tend to have suppressor effects on T-cell proliferation to mycobacterial antigens. Susceptibility and resistance to *Mycobacterium tuberculosis* in murine models depend upon a number of genes, including *sst1* (*s*usceptibility to *t*uberculosis *1*) and *Nramp1* (*n*atural *r*esistance-*a*ssociated *m*acrophage *p*rotein *1*). At least 11 polymorphisms have been identified in the human *Nramp1* homolog and studies are underway to link individual polymorphisms to susceptibility.

Where the host has difficulty in effectively eliminating these organisms, the chronic CMI response to local antigen leads to the accumulation of densely packed macrophages which release angiogenic and fibrogenic factors and stimulate the formation of granulation tissue and ultimately fibrosis. The activated macrophages, perhaps under the stimulus of IL-4, transform to epithelioid cells and fuse to become giant cells. As suggested earlier, the resulting granuloma represents an attempt by the body to isolate a site of persistent infection.

Leprosy

In human leprosy, the disease presents as a spectrum ranging from the **tuberculoid** form, with lesions containing small numbers of viable organisms, to the **lepromatous** form, characterized by an abundance of *M. leprae* within the macrophages. The tuberculoid state is associated with good cell-mediated dermal hypersensitivity reactions and a bias towards Th1-

type responses, although these are still not good enough to eradicate the bacilli completely. In the lepromatous form, there is poor T-cell reactivity to whole bacilli and poor lepromin dermal responses, although there are numerous plasma cells which contribute to a high level of circulating antibody and indicate a more prominent Th2 activity. Clearly, CMI rather than humoral immunity is important for the control of the leprosy bacillus, but reasons for the inadequate responses of the lepromatous patients are still uncertain.

IMMUNITY TO VIRAL INFECTION

Genetically controlled constitutional factors which render a host or certain of their cells nonpermissive (i.e. resistant to takeover of their replicative machinery by virus) play a dominant role in influencing the vulnerability of a given individual to infection. Macrophages may readily take up viruses nonspecifically and kill them. However, in some instances, the macrophages allow replication and, if the virus is capable of producing cytopathic effects in various organs, the infection may be lethal; with noncytopathic agents, such as lymphocytic choriomeningitis, Aleutian mink disease and equine infectious anemia viruses, a persistent infection may result. Viruses can avoid recognition by the host's immune system by latency or by sheltering in privileged sites, but they have also evolved a maliciously cunning series of evasive strategies.

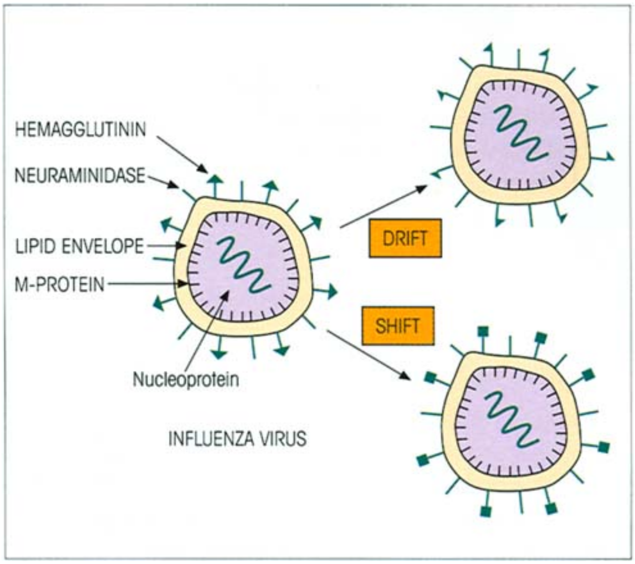

Figure 13.18. Antigenic drift and shift in influenza virus. The changes in hemagglutinin structure caused by drift may be small enough to allow protection by immunity to earlier strains. This may not happen with radical changes in the antigen associated with antigenic shift and so new virus epidemics break out. There have been 31 documented influenza pandemics (widespread epidemics occurring throughout the population) since the first well-described pandemic of 1580. In the last century there were three, associated with the emergence by antigenic shift of the Spanish flu in 1918 with the structure H_1N_1 (the official nomenclature assigns numbers to each hemagglutinin and neuraminidase major variant), Asian flu in 1957 (H_2N_2) and Hong Kong flu in 1968 (H_3N_2); note that each new epidemic was associated with a fundamental change in the hemagglutinin. The pandemic in 1918 killed an estimated 40 million people.

Immunity can be evaded by antigen changes

Changing antigens by drift and shift

In the course of their constant duel with the immune system, viruses are continually changing the structure of their surface antigens. They do so by processes termed 'antigenic drift' and 'antigenic shift', the nature of which may be made more apparent by consideration of different influenza strains. The surface of the influenza virus contains a hemagglutinin, by which it adheres to cells prior to infection, and a neuraminidase, which releases newly formed virus from the surface sialic acid of the infected cell; of these, the hemagglutinin is the more important for the establishment of protective immunity. Minor changes in antigenicity of the hemagglutinin occur through point mutations in the viral genome (**drift**), but major changes arise through wholesale swapping of genetic material with reservoirs of different viruses in other animal hosts (**shift**) (figure 13.18). When alterations in the hemagglutinin are sufficient to render previous immunity ineffective, new influenza epidemics break out.

Mutant forms can be favored by selection pressure from antibody. In fact, one current strategy for generating mutants in a given epitope is to grow the virus in tissue culture in the presence of a monoclonal antibody which reacts with that epitope; only mutants which do not bind the monoclonal will escape and grow out. This principle underlies the antigenic variation characteristic of the common cold rhinoviruses. The site for attachment and penetration of mucosal cells bearing ICAM-1 is a hydrophobic pocket lying on the floor of a canyon which antibodies were thought to be too large to penetrate. Many antibodies react with the rim of the viral canyon and mutations in the rim would thus enable the virus to escape from the host immune response without affecting the conserved site for binding to the target cell. However, one recent structural study identified three neutralizing antibodies which bear Fabs that contact a significant proportion of the canyon directly overlapping with the ICAM-1-binding site. Hydrophobic drugs have been synthe-